

Introduction

- The gastrointestinal tract has many functions such as digestion, absorption and excretion as well as the synthesis of many hormones.
- The enteric nervous system is present in the gut to control its function and communicate with the central and peripheral nervous systems.
- The gastrointestinal tract by the large foreign antigens that contains share in the expression of the immune response.

Gastrointestinal symptoms and signs:

- Many of the gastrointestinal consultations are for non-organic symptoms (functional).
- As 20% of all cancers occur in the gastrointestinal tract, therefore the clinician's main task is to recognize an organic disease among the symptoms, signs and investigations.
- In developing countries, malnutrition, viral and bacterial infections and parasitic infestations further add to the picture.
- The physician must sought for '**Alarm**' **symptoms- Red flag symptoms:**
 - Dysphagia.
 - Weight loss.
 - persistent vomiting.
 - Anorexia.
 - Hematemesis or melena.
- Patients with these features have a higher possibility of significant gastrointestinal pathology and should be investigated properly.

Stomatitis:

- It is an inflammation in the mouth from any cause (e,g: ill-fitting dentures).
- Angular stomatitis is inflammation of the corners of the mouth .

Dysphagia: Difficulty of swallowing.

Odynophagia: Painful swallowing.

Dyspepsia:

It is an inexact term used to describe a number of upper abdominal symptoms such as heartburn, acidity, pain or discomfort, nausea, wind, fullness or belching.

Vomiting:

Vomiting centers are located in the reticular formation in the medulla and are stimulated by the chemoreceptor trigger zone (CTZ), and by the vagus.

Direct stimulation of these zones also occurs by drugs, motion sickness and metabolic causes.

Two phases will be experienced by the patient before the actual vomiting :

Nausea: associated with autonomic symptoms such as salivation , pallor , sweating.

Retching: strong , involuntary , unproductive effort of vomiting associated with contraction of abdominal muscles.

The amount and character of vomiting may suggest the pathology:

- large in amount and projectile may suggest gastric outlet obstruction or upper intestinal obstruction.
- Feculent vomiting may suggest lower intestinal obstruction.

Causes of vomiting:

-Any Gastrointestinal disease

-Infections (viral and bacterial)

-CNS diseases

- Increase Intracranial pressure
- Vestibular disease
- Migraine

-Metabolic causes : Chronic renal failure , diabetic Ketoacidosis.

-Drugs : e.g Digoxin.

-Reflex : e.g. Myocardial infarction.

-Psychogenic

-Pregnancy

-Alcohol

Flatulence

This term describes excessive wind.

It is used to indicate belching, abdominal distension, gurgling and the passage of flatus per rectum.

Among causes of increase gas production are colonic bacterial breakdown of non-absorbed carbohydrate, high fiber diet , carbonated drinks.

Diarrhea and constipation:

- Common complaint in the community and not necessarily due to organic disease.

Diarrhea:

It can be defined as change of normal bowel habits in the form of

- Increase in frequency.
- Increase in amount.
- Decrease in consistency.

Feces exceeding 200g per day when dietary fiber content is low is considered diarrhea

- The presence of large volume of watery stool is usually due to organic cause.
- Bloody diarrhea is usually due to a colonic cause.
- The presence of fever and acute diarrhea for several days is usually due to an infective cause .
- A single episode of diarrhea is usually due to dietary changes or may be psychogenic.

Constipation:

The diagnosis of constipation is difficult as some individuals thought that they must open their bowel on daily basis , the presence of hard stool can be considered also a constipation

Constipation can be defined as the presence of 2 or more of the following:

- Infrequent Stools <3/week.
- Hard Stools >1/4 times of defecation.
- Incomplete evacuation >1/4 times of defecation.
- Straining in >1/4 times of defecation.
- Manual maneuvers as digital evacuation and support of pelvic floor >1/4 times of defecation.

Abdominal pain:

Pain is stimulated mainly by the stretching of smooth muscle or organ capsules.

Severe acute abdominal pain can be due to a large number of gastrointestinal conditions, and normally presents as an emergency.

An apparent 'acute abdomen' can occasionally be due to referred pain from the chest, as in pneumonia or to metabolic causes, such as diabetic ketoacidosis or porphyria.

Check:

- Intensity, character, duration and frequency of the pain
- Aggravating and relieving factors
- Associated symptoms, including non-gastrointestinal symptoms.

Upper abdominal pain:

Epigastric pain is very common and is often related to food intake.

Although functional dyspepsia is the commonest diagnosis, the symptoms of peptic ulcer disease can be identical.

Heartburn (a burning pain behind the sternum) is a common symptom of gastro-esophageal reflux.

Right hypochondrial pain:

may originate from the gall bladder or biliary tract.

Biliary pain can also be epigastric.

Biliary pain is typically intermittent and severe, lasts a few hours and remits spontaneously to recur weeks or months later.

Hepatic congestion (e.g. in hepatitis or cardiac failure) and sometimes peptic ulcer disease can present with pain in the right hypochondrium.

Chronic, persistent or constant pain in the right (or left) hypochondrium in a well-looking patient is a frequent functional symptom.

Lower abdominal pain:

Pain in the left iliac fossa may be colonic in origin (e.g. acute diverticulitis) but chronic pain is most commonly associated with functional bowel disorders.

Lower abdominal pain in women occurs in a number of gynaecological disorders and the differentiation from GI disease is important, but may be difficult.

Pain in the right iliac fossa may be due to acute appendicitis or ilececal disease, but may also be functional

Abdominal wall pain:

Persistent abdominal pain with localized tenderness, which is not relieved by tensing the abdominal muscles.

Anorexia and weight loss

Anorexia describes reduced appetite.

It is common in systemic disease and may be seen in psychiatric disorders.

Anorexia often accompanies cancer but is usually a late symptom and not of diagnostic help.

Weight loss is almost always due to reduced food intake and is a frequent accompaniment of gastrointestinal diseases.

Weight loss in malabsorption disorders is primarily due to anorexia.

Weight loss with a normal or increased dietary intake may occur with hyperthyroidism and other catabolic states.

Weight loss should always be assessed objectively as patients' impressions are unreliable.

Gastrointestinal (GIT) bleeding:

It presents with one of the followings:

Hematemesis (the vomiting of blood)

Melena (the passage of black tarry stools)

- It can occur with bleeding from any lesion proximal to the right colon.
- It needs more than 50 ml blood .
- The black color due to blood altered by passage through the gut.
- Following a bleed from the upper G I tract, unaltered blood can appear per rectum, but the bleeding must be massive and is almost always accompanied by shock.
- The passage of dark blood and clots without shock is always due to lower GI bleeding.

Hematochezia :

It is bright red or maroon rectal bleeding.

It usually occurs with lesion distal to the ligament of Treitz (at junction between duodenum and jejunum) but it can occur with massive upper GIT bleeding with rapid transit time.

Investigations

Routine haematology and biochemistry, followed by endoscopy and radiology, are the principal investigations.

Laboratory

Stool examination is useful to confirm a patient's complaint (e.g. passing of blood or Steatorrhea).

Fecal occult blood test: Can detect blood in stool that is not visibly apparent

The new Fecal Immunochemical test (FIT) is more accurate and have fewer false positive tests

Fecal calprotectin: when elevated in stool, it indicates intestinal inflammation, calprotectin increase is seen with inflammatory bowel disease

Tumour markers of the gastrointestinal tract include CEA for colon cancer, CA 19-9 for pancreatic cancer, also AFP for hepatocellular cancer

The Endoscopy

Esophagogastroduodenoscopy (EGD)

It is the investigation of choice for upper GI disorders with the possibility of therapy and mucosal biopsy.

Findings include reflux esophagitis, gastritis, ulcers and cancer

Therapeutic EGD is used to treat upper GI haemorrhage and both benign and malignant obstruction.

Relative contraindications include severe chronic obstructive pulmonary disease, a recent myocardial infarction.

Sigmoidoscopy :

Can be considered as part of the routine hospital examination in cases of diarrhea and in patients with lower abdominal symptoms such as a change in bowel habit or rectal bleeding.

The rigid sigmoidoscope allows inspection of a maximum of 20-25 cm of distal colon.

Flexible sigmoidoscope can reach up to the splenic flexure

Colonoscopy

It allows good visualization of the whole colon and terminal ileum.

Biopsies can be obtained and polyps removed.

Benign strictures can be dilated and malignant strictures, stented Cancer, polyps and diverticular disease are the commonest significant findings.

Balloon enteroscopy

Either double or single balloon, can examine the small bowel from the duodenum to the ileum using specialized enteroscopes in expert centers.

Capsule endoscopy

Used for the evaluation of obscure GI bleeding (after negative gastroscopy and colonoscopy) and for the detection of small bowel tumors and occult inflammatory bowel disease.

It should be avoided if strictures are suspected.

Endoscopic ultrasound (EUS)

Performed with a gastroscope incorporating an ultrasound probe at the tip.

It is used diagnostically for lesions in the esophageal or gastric wall, including the TNM staging of oesophageal/gastric cancer and for the detection and biopsy of pancreatic tumors and cysts.

Imaging

Full clinical information must be provided before the examination, and ideally, the images obtained should be reviewed with the radiologist to aid interpretation.

The optimal technique to be used will depend on local expertise.

1. Plain X-rays of the chest and abdomen

Are chiefly used in the investigation of an acute abdomen.

Interpretation depends on analysis of gas shadows inside and outside the bowel.

Plain films are particularly useful where obstruction or perforation is suspected, to exclude toxic megacolon in colitis and to assess fecal loading in constipation.

Calcification may be seen with gall bladder stones and in chronic pancreatitis, though CT is more sensitive for both.

2. Ultrasound

involves no radiation and is the first-line investigation for abdominal distension, e.g. ascites, mass or suspected inflammatory conditions.

It can show dilated fluid-filled loops of bowel in obstruction and thickening of the bowel wall.

It can be used to guide biopsies or percutaneous drainage.

In an acute abdomen, ultrasound can diagnose cholecystitis, appendicitis, enlarged mesenteric glands and other inflammatory conditions.

3. Computed tomography (CT)

With intraluminal contrast injection (e.g gastrografin) and intra-venous contrast injection enhance diagnostic capability (kidney functions must be assessed before the IV injection of the contrast).

CT is widely used as a first line investigation for the acute abdomen.

CT is sensitive for small volumes of gas from a perforated viscus as well as leakage of contrast from the gut lumen.

Inflammatory conditions such as abscesses, appendicitis, diverticulitis, Crohn's disease and its complications are well demonstrated.

In high-grade bowel obstruction, CT is usually diagnostic of both the presence and the cause of the obstruction.

CT is widely used in cancer staging and as guidance for biopsy of tumor or lymph nodes.

CT colonography (virtual colonoscopy) after CO₂ insufflation into a previously cleansed colon provides an alternative to colonoscopy for diagnosis of colon mass lesions .

4. Magnetic Resonance Imaging MRI

It has the advantage over CT that it uses no radiation and is useful in the evaluation of rectal cancers and abscesses and fistula in the perianal region.

It is also useful in small bowel disease and in hepatobiliary and pancreatic disease.

5. Magnetic resonance enterography

Is a magnetic resonance imaging technique used to evaluate bowel wall features of both upper and lower gastro-intestinal tract.

Although it is usually used for small bowel evaluation.

It is a less invasive technique with the advantages of no ionizing radiation exposure, multiplanarity and high contrast resolution for soft tissue.

6. Positron emission tomography (PET)

relies on detection of the metabolism of fluorodeoxyglucose.

It is used for staging esophageal, gastric and colorectal cancer and in the detection of metastatic and recurrent disease

7. Contrast studies

Barium swallow:

Examines the esophagus and proximal stomach.

Its main use is for investigating dysphagia.

Double-contrast barium meal:

Examines the esophagus, stomach and duodenum, however Gastroscopy is a more sensitive test and enables biopsy of suspicious areas.

Small bowel follow-through:

Specifically examines the fold pattern and calibre of the small bowel .

Specific views of the terminal ileum can be obtained and are used to identify early changes in patients with suspected Crohn's disease or TB

Small bowel enema (enteroclysis):

A tube is passed into the duodenum and a large volume of dilute barium is introduced.

It is particularly used to demonstrate strictures or adhesions when there is suspicion of intermittent obstruction.

Generally, this has been replaced by MR enteroclysis.

Barium enema examines the colon and is used for altered bowel habit.

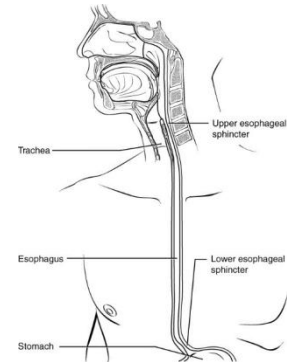
Colonoscopy and CT colonography have largely replaced this examination for rectal bleeding, polyps and inflammatory bowel disease.

Absorbable water-soluble (Gastrografin) contrast agents should be used in preference to barium when perforation is suspected anywhere in the gut.

The esophagus

Structure and physiology:

- the esophagus is 20 cm long that connects the pharynx to the stomach.
- the esophagus is lined by stratified squamous epithelium.
- the esophagus is separated from the pharynx by the upper esophageal sphincter (UES).
- The lower esophageal sphincter (LES) consists of a 2-4 cm in the distal end of the esophagus that responsible for the prevention of gastric reflux.



Symptoms of esophageal disorders

1. Dysphagia:

- Difficulty of swallowing.
- The characteristics of the progression of dysphagia can be helpful progression over a few weeks suggests a malignant stricture.
- The slow onset of dysphagia for solids and liquids at same time suggest motility disorder (e.g. Achalasia).

Causes of dysphagia

- Disease of mouth and tongue e.g. Tonsillitis .
- Neuromuscular disorders e.g. Bulbar palsy , myasthenia gravis .
- Esophageal motility disorders e.g. Achalasia ,scleroderma ,diffuse esophageal spasm, diabetes mellitus, chagas' disease.
- Extrinsic pressure e.g. Mediastinal glands , goiter ,enlarged left atrium
- Intrinsic lesion e.g. Foreign body, stricture: Benign or malignant - carcinoma, esophageal web ,pharyngeal pouch.

2. Odynophagia:

Is pain during swallowing and is suggestive of esophagitis.

3. Substernal discomfort, heartburn:

This is a common symptom of reflux of gastric contents into the esophagus.

4. Regurgitation:

Is the effortless reflux of esophageal contents into the mouth and pharynx.

Signs of esophageal disorders

The main sign of esophageal disease is weight loss due to reduced food intake.

Investigations for esophageal disorders

1. Esophagoscopy.

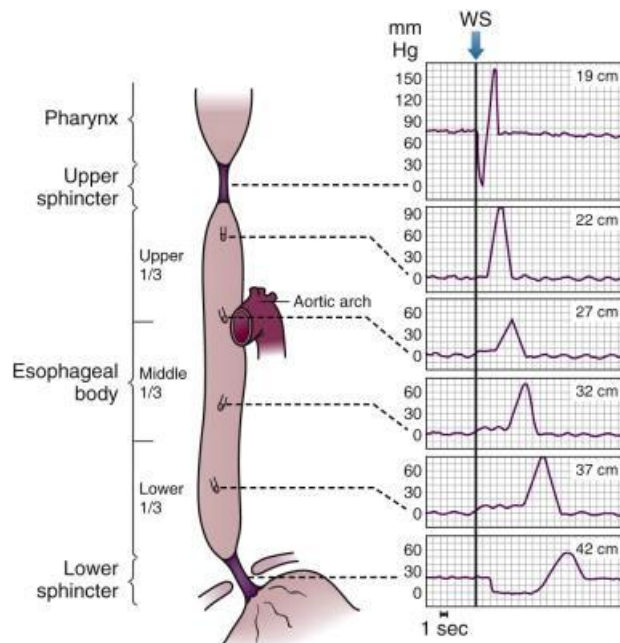
2. Barium swallow and meal.

3. Manometry:

It is used to assess esophageal motor activity.

4. PH monitoring:

24-hour ambulatory monitoring uses a PH-sensitive probe positioned in the lower esophagus



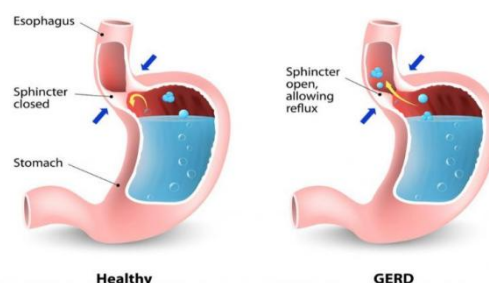
Gastroesophageal reflux disease (GERD)

Pathophysiology

- The reflux occurs when the anti-reflux mechanisms fail, allowing acidic gastric contents to make prolonged contact with the lower esophageal mucosa.
- Decrease tone of lower esophageal sphincter (LES).

Risk factors

- 1) Increase intra-abdominal pressure : Pregnancy or obesity
- 2) Diet: Fat, chocolate, coffee, alcohol and large meals.
- 3) Drugs: Antimuscarinic, calcium channel blockers, nitrates.
- 4) Cigarette smoking
- 5) Hiatus hernia (described below)



Clinical features

- **Heartburn** is the major feature.
The burning is aggravated by lying down and relieved by oral antacids.
- **Regurgitation** of food and acid into the mouth.
- **Aspiration pneumonia**

Diagnosis and investigations

- The clinical diagnosis can usually be made without investigation.
- If there is **alarm signs** e.g. Weight loss, no response to ppi the following Investigations:
 - 1) 24-hour pH monitoring.
 - 2) Upper endoscopy.
 - 3) Manometry.

Esophageal reflux and cardiac ischemia

- Reflux pain: Burning, worse on lying down , don't radiates to the arms ,worse with hot drinks or alcohol and is relieved by antacids
- Cardiac ischemic pain: Gripping or crushing radiates to neck or left arm ,worse with exercise accompanied by dyspnea

Treatment

- Simple measures :

- Simple antacids.
- Loss of weight and raising the head of the bed at night.
- Reduction in caffeine consumption.
- Stop smoking and alcohol intake.

- Drugs:

- Proton pump inhibitors e.g. Omeprazole, reduce gastric acid secretion and are the drugs of choice for mild cases.
- H2 blockers e.g. Ranitidine.
- Antacids.
- Dopamine antagonist as metoclopramide and domperidone are increase peristalsis and speed emptying .

- Endoluminal gastroplication:

This recent technique could be performed by endoscopy:

■ endoscopic full-thickness plication (GERDX):

A endoscopic device has been designed to inhibit gastroesophageal reflux by placing a transmural plication near the gastroesophageal junction under direct endoscopic visualization.

■ Anti-reflux mucosal ablation (ARMA):

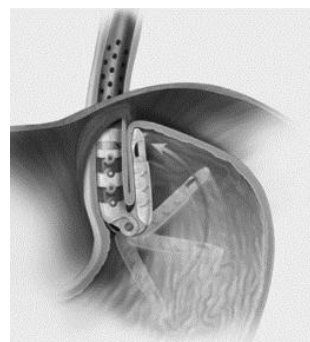
Mucosal ablation around the gastroesophageal junction to cause a healing process to re-form the cardiac opening

- Surgery:

Management of sliding hiatus hernia (described below)

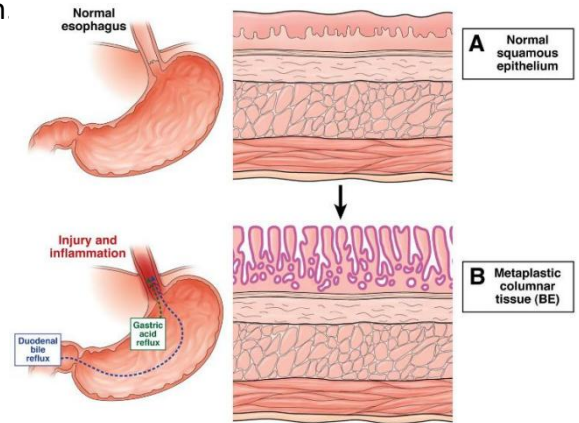
Complications

- peptic stricture: since the introduction of PPIs, peptic strictures have become less common.



Barrett's esophagus

- The normal esophageal squamous epithelium is replaced by metaplastic columnar mucosa.
- Premalignant: Esophageal adenocarcinoma.
- Barrett's is commonest in middle-aged obese men.
- Diagnosis: Endoscopy, chromo-endoscopy.
- Treatment : Radiofrequency ablation (RFA).



Benign esophageal stricture

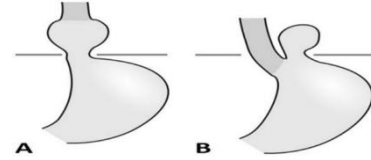
- Peptic stricture secondary to reflux is the most common cause of benign strictures
- They also occur after the ingestion of corrosives, after radiotherapy, after sclerosis of varices.
- Dysphagia is usually managed by endoscopic dilatation.
- Surgery is sometimes required.

Esophageal infections

- Infection is a cause of painful swallowing and is seen in immunosuppressed (e.g. On Chemotherapy) and aids.
- Infection can occur with:
 - candida - herpes simplex - cytomegalovirus - tuberculosis.
- In candidiasis, the characterize by white plaques.
- Most patients on large doses of immunosuppressive agents are treated prophylactically for candida with nystatin or amphotericin.

Hiatus hernia

- A hiatus hernia describes the protrusion of an organ from the abdominal cavity into the chest through the esophageal hiatus.
- This is typically the stomach that herniates.
- Extremely common disease.



Classification

Sliding hiatus hernia (85 %): The gastro-oesophageal Junction (GOJ), the abdominal part of the oesophagus, and the cardia of the stomach “slide” upwards through the diaphragmatic hiatus into the thorax. (fig. A)

Rolling or paraesophageal hernia (10%): An upward movement of the gastric fundus through esophageal hiatus occurs to lie alongside a normally placed GOJ. This is a true hernia with a peritoneal sac. (fig. B)

A **mixed** type hiatus hernia: which has both a rolling and sliding components

Risk factors

Age:

Is an important risk factor, due to:

A combination of age- related loss of diaphragmatic tone, increasing intra-abdominal pressures (e.g. Repetitive coughing), and widening of diaphragmatic hiatus.

Increased intra-abdominal pressure:

Pregnancy, obesity, and ascites due to upwards displacement of the abdominal viscera.

Clinical features of sliding h.h

The majority of hiatus hernias are **asymptomatic**.

Gastro-esophageal reflux symptoms:

- 1) Marked **burning epigastric pain**: made worse by lying flat and on leaning forwards, patient experiences relief on raising head of bed.
- 2) **Vomiting – Regurgitation**.
- 3) **Bleeding** and **anemia** secondary to esophageal ulceration leading either melena, occult blood in stool or hematemesis in severe cases.
- 4) **Dysphagia** (esophageal stricture on long-standing cases)
- 5) Sometimes **weight loss** in case of long duration vomiting and dyspepsia.

Epigastric tenderness in case of associated gastritis

Investigations

1. upper endoscopy:

It is the standard investigation for esophageal disorders showing upward displacement of the gastro-esophageal junction also (z-line).

During inspiration the GOJ is opened and closed in expiration

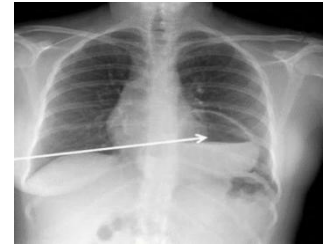
It shows evidence of reflux as hyperemia, fissuring, ulceration or stricture of the lower esophagus

Biopsy may be taken from suspicious areas (areas of barrett's)

2. Plain x-ray chest:

Fundic air may be evident in the chest

Detects chest complications as aspiration pneumonia.



CXR: stomach in the chest

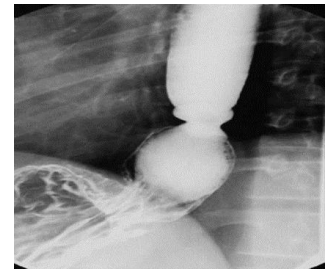
3. Barium swallow:

Evidence of complication as stricture esophagus.

4. Barium meal in **trendlenberg's** position:

Reflux of the barium to the esophagus in GERD

Stomach above the diaphragm in big hiatus hernia.



Barium meal: sliding H.H

5. Ct abdomen and chest

6. Ph test: Shows increased acid level in the esophagus. **(diagnostic for reflux)**

Treatment:

Life-style changes as:

- Losing weight.
- Decreasing the size of meals.
- Avoiding certain acidic foods, such as tomato sauce and citrus
- Limiting fried and fatty foods, foods containing caffeine (including chocolate),
- Eating meals at least three to four hours before lying down, and avoiding bedtime snacks
- Keeping head higher than the rest of body when lying on the back.
- Stop smoking
- Avoid a tight belt or tight clothing.

Medications:

- I. Antacids that neutralize stomach acid.
- li. H₂ blockers.
- lii. Proton pump inhibitors.

NB: ppi stronger acid blockers than h₂ blockers and allow for damaged esophageal tissue to heal.

Surgery:

- To correct gastro-esophageal reflux by creating a competent valve mechanism at the lower end of the esophagus.
- To repair the widened esophageal hiatus

Indications:

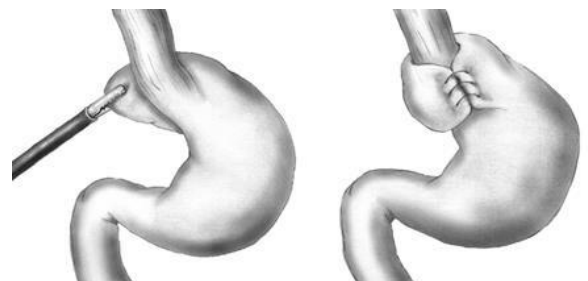
- severe symptoms of reflux with failure of medical treatment
- patient has large h.h.
- presence of complications: (ulceration, severe inflammation, bleeding, stricture, Barrett's)

Fundoplication:

- Surgery involves pulling the hiatal hernia (fundus) back into the abdomen
- Wrapping the fundus of the stomach around the lower portion of the esophagus.
- This creates a permanently competent sphincter so that stomach contents will not reflux back into the esophagus
- Repair the wide esophageal hiatus of the diaphragm.
- The fundic wrap is 360° (in nissen's technique) and 270° in (toupet technique)
- The procedure can be done laparoscopically better than laparotomy.

Complications of fundoplication:

- O Dysphagia
- O Inability to belch
- O Slipped nissen and recurrence of herniation



Achalasia of esophagus

- It is a primary esophageal motility disorder characterized by failure of LES relaxation and loss of peristalsis in the distal esophagus.

Pathophysiology

Vagal degeneration:

LES contraction and relaxation are regulated through vagus nerve

Degeneration of ganglion cells in mesenteric plexus concerned with the inhibitory neurons → increased tone in LES → severe spasm.

Causes of achalasia

I- primary achalasia: Etiology unknown, hereditary, degenerative, autoimmune

II-secondary achalasia: (e.g. Secondary to **chaga's** disease)

Clinical presentations

- Dysphagia for liquids more than solids most significant symptom. (effect of weight).
- Difficulty belching.
- Regurgitation (especially when lying flat at night)
- Heartburn and chest pain in case of esophagitis
- Weight loss is minimal and late
- Hiccups

Complications associated with achalasia

- Pneumonic aspiration
- Esophageal cancer

Investigations

I- Radiographic evaluation

II- Manometric evaluation

III- endoscopic evaluation

I- radiographic evaluation

1- chest x-ray:

Markedly dilated esophagus with widening of mediastinum

Absent gastric air bubble: Swallowed air does not reach the fundus

2- barium swallows: **Diagnostic tool**

Smooth narrowing of the distal esophagus at the level of the lower esophageal sphincter with the characteristic “**bird beak**” appearance.

The proximal esophagus is markedly dilated

Absent gastric air bubble

II-esophageal manometry: Confirm the diagnosis

Elevated resting LES pressure (spasm).

Incomplete LES relaxation.

Weak peristalsis of esophageal wall (aperistalsis).

III- endoscopic evaluation:

To exclude cancer of the gastro-esophageal junction.

Detects spasm of the LES.

Management

Pharmacological agents:

Patients who are unable to tolerate more invasive treatment modalities.

1- Calcium channel blockers and nitrates decrease LES pressure.

2- Endoscopic intra-sphincteric injection of botulinum toxin:

Blocks release of Acetylcholine at the level of the LES → relaxation of the sphincter

Endoscopic techniques:

A- pneumatic dilatation:

It is the most effective non-surgical treatment option for patients with achalasia.

The balloon is inflated at the level of the gastro-esophageal junction to dilate the muscle fibers while leaving the mucosa intact.

B- The per-oral endoscopic myotomy, or **Poem**:

It is a minimally invasive endoscopic procedure for the treatment of achalasia

Where in the inner circular muscle layer of the lower esophageal sphincter is divided through a submucosal tunnel.

Surgical:

Laparoscopic heller's cardio-myotomy

Weakens LES by cutting muscle fibers through performing an anterior myotomy across the LES.

The Stomach and duodenum

Anatomy

The stomach consists of:

Cardia: adjacent to the gastroesophageal junction.

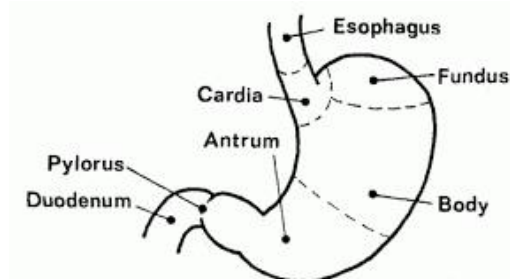
Fundus: dome-shaped and extends above and to the left of the cardia, toward the left hemidiaphragm.

Body: extends from the fundus to the lower end of the lesser curve, known as the incisura angularis.

Antrum: extends from the incisura to the pyloric canal.

Pylorus: area between antrum and duodenum.

- There are two sphincters, the gastro-oesophageal sphincter and the pyloric sphincter.
- The mucosa of the upper two-thirds of the stomach contains parietal cells that secrete hydrochloric acid, and chief cells that secrete pepsinogen (which initiates proteolysis).
- The antral mucosa secretes bicarbonate and contains mucus-secreting cells and G cells, which secrete gastrin, stimulating acid production.
- Somatostatin, a suppressant of acid secretion, is also produced by specialized antral cells (D cells).
- Mucus-secreting cells are present throughout the stomach and secrete mucus and bicarbonate.
- The mucus is made of glycoproteins called mucins.
- The 'mucosal barrier', made up of the plasma membranes of mucosal cells and the mucus layer, protects the gastric epithelium from damage by acid, aspirin, NSAIDs alcohol and bile salts.
- Prostaglandins stimulate secretion of mucus, and their synthesis is inhibited by aspirin and NSAIDs, which inhibit cyclo-oxygenase
- The duodenal mucosa has villi like the rest of the small bowel, and also contains Brunner's glands that secrete alkaline mucus, This along with the pancreatic and biliary secretions, helps to neutralize the acid secretion from the stomach when it reaches the duodenum.



Physiology

Acid secretion is central to the functionality of the stomach.

Acid is not essential for digestion but does prevent some food-borne infections.

It is under neural and hormonal control and both stimulate acid secretion through the direct action of histamine on the parietal cell.

Acetylcholine and gastrin also release histamine via the enterochromaffin cells.

Somatostatin inhibits both histamine and gastrin release and therefore acid secretion.

Other major gastric functions are:

- Reservoir for food.
- Emulsification of fat and mixing of gastric contents.
- Secretion of intrinsic factor .

Gastritis

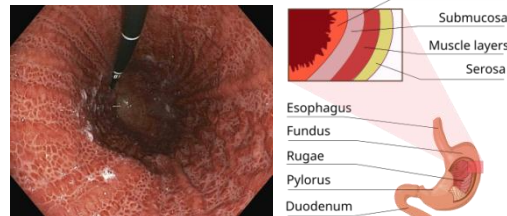
Gastritis is the inflammation of the gastric mucosa.

Gastritis is classified based on the acuity of the condition (acute versus chronic), the histological features of inflammation, or the etiology.

Gastropathy is a gastric mucosal disorder without inflammation, featuring epithelial injury and subsequent regeneration.

Gastropathy is usually caused by:

- Bile reflux.
- Hypovolemia.
- Chronic congestion as in portal hypertension (called portal hypertensive gastropathy)



Gastritis and gastropathy are not mutually exclusive conditions and might sometimes coexist.

In clinical practice, gastritis may be accompanied by signs of mucosal injury, whereas gastropathy may present with an inflammatory reaction in the gastric mucosa.

Causes of Gastritis :

- 1-H.pylori gastritis (commonest cause).
- 2-Erosive and hemorrhagic gastritis (NSAIDs).
- 3-Autoimmune gastritis.
- 4-Viral infection (CMV & herpes simplex).
- 5-Crohn's disease.

Acute Gastritis

Acute gastritis refers to a sudden onset of inflammation of the gastric mucosa.

In contrast, chronic gastritis refers to long-lasting inflammation of the gastric mucosa.

With acute gastritis, a disruption in the gastric mucosa triggers an inflammatory immune response that attracts white blood cells to the site of injury.

If the mucosal damage is severe enough, acute gastritis can progress to erosive gastritis, which consists of shallow lesions of the mucosa (i.e. **gastric erosions**), ulcerations , and small areas of bleeding within the mucosa.

Causes of acute gastritis

Acute gastritis occurs as a result of weakness or injury to the gastric mucosa, which can allow stomach acids to further damage and inflame the lining.

Frequent or long-term use of non-steroidal anti-inflammatory drugs (**NSAIDs**), such as aspirin and ibuprofen. (**Erosive and hemorrhagic gastritis**)

NSAIDs work by blocking the enzyme **cyclo-oxygenase**, resulting in inhibition of production of **prostaglandin E2** that normally increase the production of gastric mucus and decrease the production of gastric acid.

Bacterial infection by *Helicobacter pylori* (**H. pylori**) is **the commonest cause** of gastritis, Although early infection by *H. pylori* does not cause many symptoms and usually goes unnoticed. However, long-lasting or chronic infection by *H. pylori* can lead to persistent inflammation of the gastric mucosa , gastric ulcer and duodenal ulcer.

Acute gastritis can also occur as a result of extreme physiological stress, often due to major surgery, trauma, severe burns, or severe illnesses.

Additional risk factors for developing acute gastritis include increased intake of alcohol and caffeine and exposure to cigarette smoke, all of which can irritate the gastric mucosa.

Pathogenesis

Acute gastritis usually lasts for a short period of time. In most cases, it resolves spontaneously within a few days or weeks once the inflammation has settled.

In other cases, however, acute gastritis can lead to recurrent or long-term inflammation of the gastric mucosa and can lead to chronic gastritis.

With time, chronic gastritis can increase the risk of developing other complications, such as stomach ulcers, upper gastrointestinal bleeding, and certain types of stomach neoplasm

Signs and symptoms of acute gastritis

Most individuals with acute gastritis are **asymptomatic** or experience mild symptoms, such as loss of appetite, upper abdominal discomfort, belching, nausea, and vomiting.

In more severe cases, some individuals may experience upper gastrointestinal bleeding due to gastric erosions and ulcerations of the mucosa.

Diagnosis of acute gastritis

Initially, an assessment of the individual's medical history is performed to identify possible causes of acute gastritis, such as long-term use of NSAIDs, excessive alcohol consumption, or *H. pylori* infection.

Since acute gastritis is often self-resolving, if a clear cause of inflammation can be identified and treated successfully, no additional tests may be required.

However, if the diagnosis is uncertain, or if bleeding occurs, an **upper endoscopy** may be performed to take a direct look at the gastric mucosa and obtain a biopsy sample to examine.

Treatment of acute Gastritis

Treatment of acute gastritis is aimed at relieving the underlying cause of inflammation.

Some cases of acute gastritis resolve without any treatment, while others require treatment via lifestyle changes or medications.

Proton-pump inhibitors, such as omeprazole or pantoprazole, may aid in relieving the symptoms of gastritis by helping decrease gastric acid production.

Proton-pump inhibitors, along with a combination of various **antibiotics**, can also be used to treat *H. pylori* infection.

Along with proton-pump inhibitors, medication may feature **antacids**, including **H2 blockers** (e.g., famotidine).

It is also important, when possible, to remove or avoid any risk factors that may contribute to the inflammation.

This may include ceasing use of NSAIDs, reducing caffeine intake, stopping alcohol intake and smoking.

Chronic Gastritis

There are two types of chronic gastritis: chronic autoimmune gastritis and *Helicobacter pylori*-associated gastritis.

Chronic autoimmune gastritis

Affects the **body** and **fundus** of the stomach (**pangastritis**) ; and it represents an example of a type **IV hypersensitivity**.

In chronic autoimmune gastritis, there are auto-antibodies against **H⁺/K⁺ ATPase** on parietal cells and intrinsic factor.

leading to **atrophic gastritis** and loss of parietal cells with **achlorhydria** and intrinsic factor deficiency lead to decreased absorption of vitamin B12 causing the clinical syndrome of **pernicious anemia**.

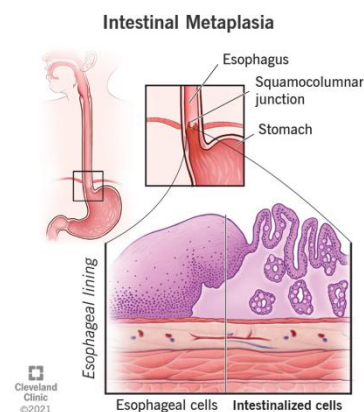
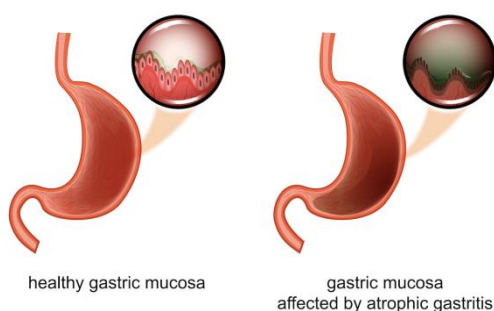
Autoantibodies against H⁺/K⁺ ATPase decrease gastric acid secretion, thereby causing **hypochlorhydria**, which is associated with **impaired iron absorption** and **hyperplasia of gastrin cells**.

Metaplasia, usually of the intestinal type, is almost always in the context of atrophic gastritis

As a result, there's an increased secretion of gastrin, **hypergastrinemia**, which is a hormone that has a trophic effect on enterochromaffin-like cells, or **ECL cells**; therefore, these individuals are at risk for developing **neuroendocrine tumors**.

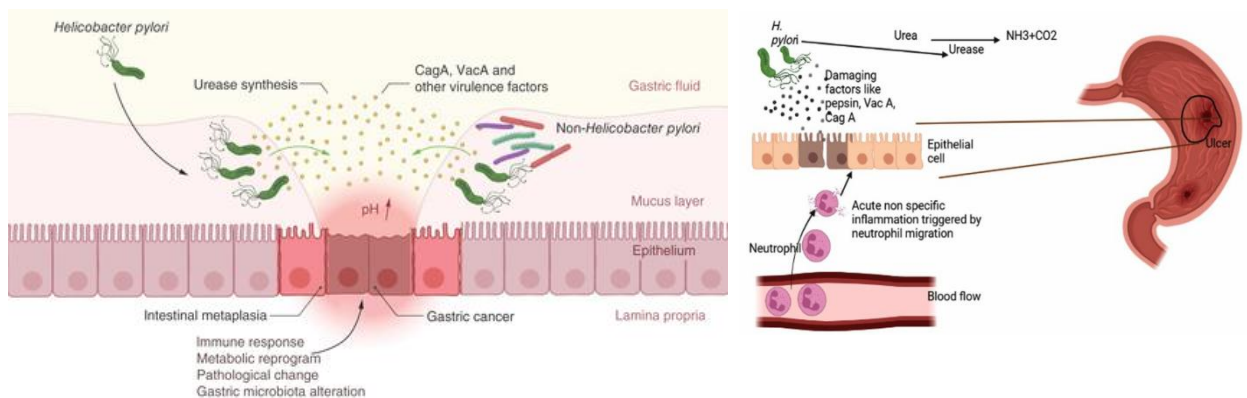
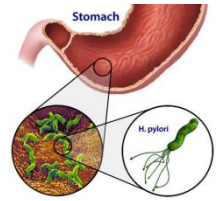
Finally, in chronic autoimmune gastritis **chief cells** are also lost and this results in **decreased levels of serum pepsinogen**.

Diagnosis of chronic autoimmune gastritis includes **endoscopic biopsy**, but also detection of **anti-parietal cell antibodies**, **anti-IF antibodies**; and **levels of serum gastrin and pepsinogen**.



Helicobacter pylori Gastritis

- *H. pylori* is a slow-growing spiral Gram-negative flagellate urease-producing bacterium which plays a major role in **gastritis** and **peptic ulcer disease**.
- It colonizes the mucous layer in the **gastric antrum**, but is also found in the duodenum in areas of gastric metaplasia.
- **Antral gastritis** is the usual effect of *H. pylori* infection. It is usually **asymptomatic**, although patients without ulcers do sometimes experience relief of dyspeptic symptoms after *Helicobacter* eradication.
- Chronic antral gastritis causes **hypergastrinaemia** due to gastrin release from antral G cells.
- The subsequent increase in acid output is usually asymptomatic, but can lead to **duodenal ulceration** *H. pylori* gastritis.



Epidemiology

The prevalence of *H. pylori* is high in developing countries (80-90% of the population), and much lower (20-50%) in developed countries.

Infection rates are highest in lower income groups.

Infection is usually acquired in childhood; although the exact route is uncertain, it may be fecal-oral or oral-oral.

The incidence increases with age.

Pathogenesis

- The pathogenetic mechanisms are not fully understood, with the majority of the colonized population remaining asymptomatic throughout their life.
- H. pylori is highly adapted to the stomach environment, exclusively colonizing gastric epithelium and inhabiting the mucous layer, or just beneath.
- The production of **urease** enables the conversion of urea to ammonium and chloride, which are directly cytotoxic.
- Ulcers are commonest when the infecting strain expresses **CagA** (cytotoxic-associated protein) and **VacA** (vacuolating toxin) genes secondary to a more pronounced inflammatory and immune response
- The precise mechanism of duodenal ulceration is unclear, as only 15% of patients infected with H. pylori (50-60% of the adult population worldwide) develop duodenal ulcers.
- Factors that have been implicated include: increased gastrin secretion, smoking, bacterial virulence and genetic susceptibility.
- **Gastric ulcer (GU):** Gastric ulcers are associated with a gastritis affecting the body as well as the antrum of the stomach (pangastritis) causing parietal cell loss and reduced acid production.
- The ulcers are thought to occur because of reduction of gastric mucosal resistance due to cytokine production by the H pylori infection or perhaps to alterations in gastric mucus.

Other H. pylori associated diseases :

- **Gastric adenocarcinoma.**
 - The incidence of distal (but not proximal) gastric cancer parallels that of H. pylori infection in countries with a high incidence of gastric cancer.
 - Serological studies show that people infected with H. pylori have a higher incidence of distal gastric carcinoma (Gastric B cell lymphoma).
 - Over 70% of patients with gastric B cell lymphomas (mucosal-associated lymphoid tissue - MALT) have H. pylori.
- H. pylori gastritis has been shown to contain the clonal B cell that eventually gives rise to the **MALT lymphoma**.

Diagnosis of Helicobacter pylori infection

Diagnosis of *H. pylori* is necessary if the clinician plans to treat a positive result.

This is usually in the context of active peptic ulcer disease, previous peptic ulcer disease or MALT lymphoma, or to 'test and treat' patients with dyspepsia under the age of 55 with no alarm symptoms (i.e. weight loss, anemia, dysphagia, vomiting, or family history of gastrointestinal cancer).

Non-invasive methods

➤ Stool antigen test.

This is beginning to be **the method of choice** to determine the *H. pylori* status.

A specific immunoassay using monoclonal antibodies for the qualitative detection of *H. pylori* antigen is now widely available.

The overall sensitivity and specificity is over 95% .

It is useful in the diagnosis of *H. pylori* infection and for monitoring efficacy of eradication therapy.

Patients should be off PPIs for 2 weeks but can continue with H2 blockers.

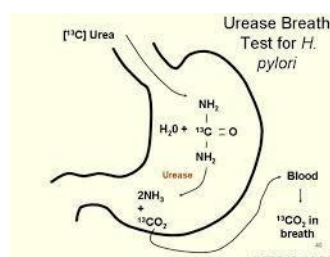
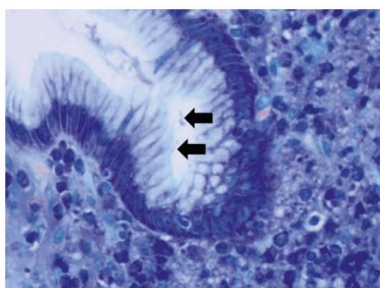
➤ ¹³C Urea breath test .

This is a quick and reliable test for *H. pylori* and can be used as a screening test.

The measurement of ¹³CO₂ in the breath after ingestion of ¹³C urea.

The test requires a mass spectrometer.

The test is **sensitive (90%) and specific (96%)**, but the sensitivity can be improved by insuring the patient has not taken antibiotics in the 4 weeks prior and proton pump inhibitors in the 2 weeks before the test.



Urea breath test for detection of *H. pylori*

➤ Serological tests detect IgG antibodies

Are reasonably sensitive (90%) and specific (83%).

They have been used in diagnosis and in epidemiological studies.

IgG titres may take up to 1 year to fall by 50% after eradication therapy and therefore are not useful for confirming eradication or the presence of a current infection.

Invasive (endoscopy)

Biopsy urease test

Gastric biopsies, usually antral are added to a substrate containing **urea and phenol red**. If *H. pylori* are present, the urease enzyme that they produce splits the urea to release **ammonia** which raises the pH of the solution and causes a rapid colour change (**yellow to red**).

This enables a patient's *H. pylori* status to be determined 28 before they leave the endoscopy unit.

The test may be falsely negative if patients are taking PPIs or antibiotics.

Histology

H. pylori can be detected histologically on routine (**Giemsa**) stained sections of gastric mucosa obtained at endoscopy.

Culture

Biopsies obtained can be cultured on a special medium, and in vitro sensitivities to antibiotics can be tested.

This technique is typically used for patients with refractory *H. pylori* infection to identify the appropriate antibiotic regimen and routine culture is rare.

- Diagnosis of *H. pylori* is necessary if the clinician plans to treat a positive result.
- This is usually in the context of active peptic ulcer disease, previous peptic ulcer disease or MALT lymphoma, or to 'test and treat' patients with dyspepsia under the age of 55y with **no alarm symptoms** i.e., weight loss, anemia, dysphagia, vomiting, or family history of gastrointestinal cancer.

Management

Eradication therapy

- One of the most commonly used regimen **triple therapy for 14 days** is :
Omeprazole 20 mg twice / day + Amoxicillin 1g twice / day + clarithromycin 500 mg twice /day . Treatment should be given for 14 days.
- Metronidazole, clarithromycin, amoxicillin, tetracycline are the most widely used agents.
- Bismuth subcitrate (120–300 mg) can be used in a quadruple therapy
- Quinolones such as ciprofloxacin are also used when standard regimens have failed ('rescue therapy').
- None of these drugs is effective alone; eradication regimens therefore usually comprise two antibiotics given with powerful acid suppression in the form of a PPI
- There is a high incidence of resistance to metronidazole and clarithromycin, particularly in some populations
- Oral metronidazole has frequent side-effects
- Current recommendations are that all patients with duodenal and gastric ulcers should have H. pylori eradication therapy if the bacteria is present.
- Eradication therapy is controversial if the patients have incidental H. pylori infection with no gastric or duodenal ulcer
- In developing countries reinfection is common, compliance with treatment may be poor and metronidazole resistance is high (>50%) as it is frequently used for parasitic infections), so failure of eradication is common.

Erosive and hemorrhagic gastritis

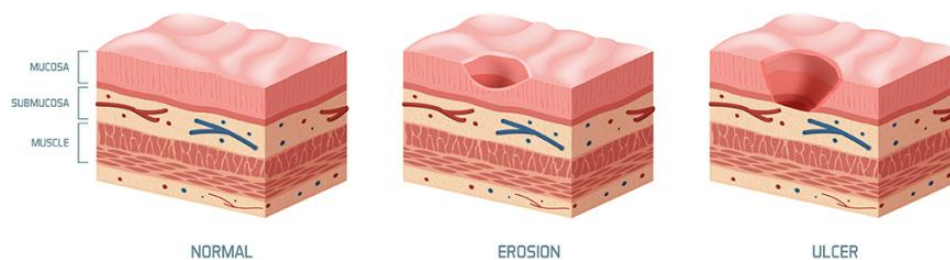
This form of gastritis is usually due to drugs as **aspirin** and **NSAIDs**.

Aspirin inhibits the synthesis of the protective prostaglandin E2 and F2 α leading to increased acid secretion and decreased formation of the protective mucosal layer.

The condition may be chronic due to recurrent use of these drugs as in patients with osteoarthritis.

Alcohol may lead to damage of the mucosal protective layer and gastric erosions.

Stress gastritis occur in stressful situations as in acute myocardial infarction and cerebrovascular strokes.



Clinical picture

- Epigastric pain
- Dyspepsia
- Gastrointestinal bleeding
- In alcoholics and chronic NSAID's users, the gastritis become chronic and may be completely asymptomatic.

Investigations

Upper endoscopy confirms the diagnosis and is mandatory in any patient having hematemesis.

Treatment

- H2 receptor blockers.
- Proton pump inhibitors.
- Sucralfate.
- When aspirin and NSAIDs should be prescribed additional gastric protective treatment may be added.

Peptic ulcer disease

- A peptic ulcer consists of a break in the superficial epithelial cells penetrating down to the muscularis mucosa of either the stomach or the duodenum; there is a fibrous base and an increase in inflammatory cells.
- **Erosions**, by contrast, are superficial breaks in the mucosa alone.
- Most DUs are found in the **duodenal cap**; the surrounding mucosa appears inflamed, hemorrhagic or friable (**duodenitis**).
- GUs are most commonly seen on the **lesser curve near the incisura**, but can be found in any part of the stomach.

Epidemiology of peptic ulcer disease:

- Duodenal ulcers affect approximately 10% of the adult population and are two to three times more common than gastric ulcers.
- Ulcer rates are declining rapidly for younger men and increasing for older individuals, particularly women.
- Both DUs and GUs are common in the elderly.
- There is considerable geographical variation, with peptic ulcer disease being more prevalent in **developing countries** related to the high H. pylori infection.
- In the developed world the percentage of NSAID-induced peptic ulcers is increasing, as the prevalence of H. pylori declines.

Zollinger Ellison syndrome is a rare cause of peptic ulcer disease which is due to a tumor called a gastrinoma.

A gastrinoma is a neuroendocrine tumor that is typically located in the duodenal wall or pancreas, and secretes abnormal amounts of gastrin.

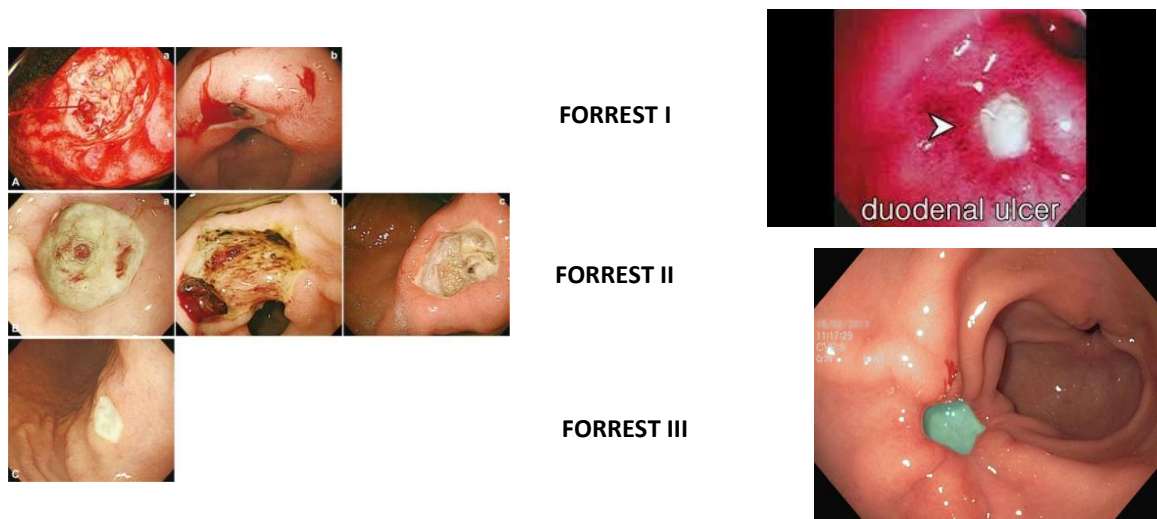
Excess gastrin stimulates parietal cells to release excess hydrochloric acid, overwhelming normal defense mechanisms and allowing ulcers to develop in the first portion of the duodenum or even in the distal duodenum or jejunum.

Clinical features of peptic ulcer disease

- The characteristic feature of peptic ulcer is recurrent, burning epigastric pain.
- It has been shown that if a patient points with a single finger to the epigastrium as the site of the pain this is strongly suggestive of peptic ulcer disease.
- The relationship of the pain to food is variable and on the whole not helpful in diagnosis.
- The pain of a DU classically occurs at night (as well as during the day) and is worse when the patient is hungry, but this is not reliable.
- The pain of both gastric and duodenal ulcers may be relieved by antacids.
- Nausea may accompany the pain; vomiting is infrequent but can relieve the pain.
- Anorexia and weight loss may occur, particularly with GUs.
- Persistent and severe pain suggests complications such as penetration into other organs.
- Back pain suggests a penetrating posterior ulcer.
- Severe ulceration can occasionally be symptomless, as many who present with acute ulcer bleeding or perforation have no preceding ulcer symptoms.
- Untreated, the symptoms of a DU relapse and remit spontaneously.
- The natural history is for the disease to remit over many years due to the onset of atrophic gastritis and a decrease in acid secretion.
- Examination is usually unhelpful ; epigastric tenderness is common in non-ulcer dyspepsia.

Investigation of suspected peptic ulcer disease

- Patients under 55 years of age with typical symptoms of peptic ulcer disease who test positive for H. pylori can start eradication therapy without further investigation.
- Endoscopic diagnosis and exclusion of cancer is required in older patients .
- All gastric ulcers must be biopsied to exclude an underlying malignancy and should be followed up endoscopically until healing has taken place.
- Endoscopy is required in all patients with alarm symptoms.
- Patients with a risk of bleeding or those with complications, i.e. haemorrhage or perforation, should always have a stool test for H. pylori or a ¹³C urea breath test 6 weeks after the end of treatment to be sure eradication is successful.



NSAIDs and ulcers

- Aspirin and other NSAIDs deplete mucosal prostaglandins by inhibiting the cyclo-oxygenase (COX) pathway, which leads to mucosal damage.
- Cyclo-oxygenase occurs in two main forms: COX-1, the constitutive enzyme, and COX-2, inducible by cytokine stimulation in areas of inflammation.
- COX-2 specific inhibitors have less effect on the COX-1 enzyme in the gastric mucosa. It still produce gastric mucosal damage but less than with other conventional NSAIDs.
- Only a small proportion of patients have symptoms (about 5%) and only 1-2% have a major problem, i.e. GI bleed.
- Because of the large number of patients on NSAIDs including low-dose aspirin for vascular prophylaxis, this is a significant problem, particularly in the elderly.

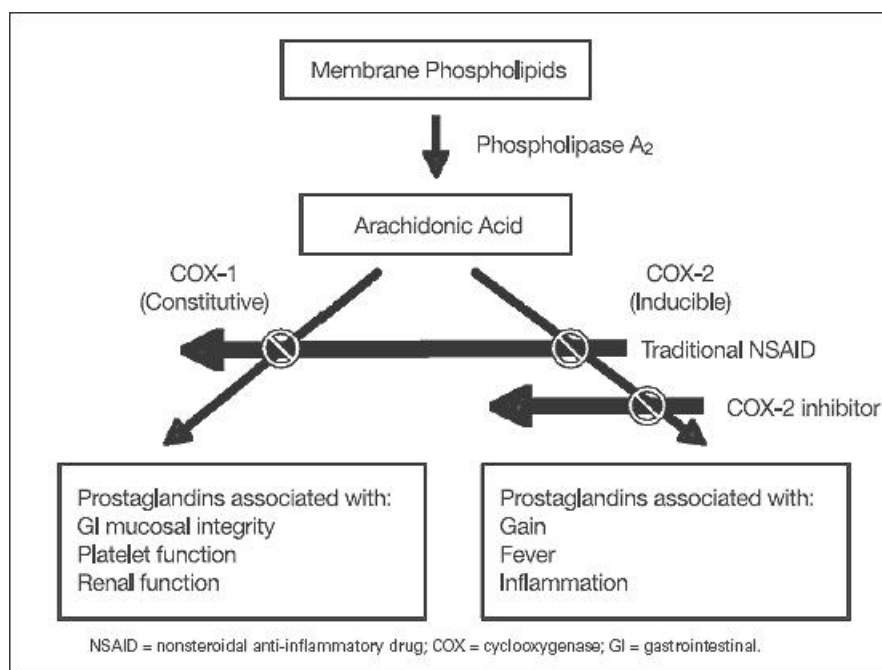


Figure 14 ; Mechanism of action of traditional NSAID and anti COX -2 inhibitors

Treatment

- Stop the ingestion of NSAIDs.
- A PPI should be given.
- H. pylori eradication therapy if H. pylori positive.
- In many people with severe arthritis, stopping NSAIDs may not be possible, therefore
 - Use: -A NSAID with low GI side-effects at lowest dose possible or if there is no cardiovascular risk, a COX-2 NSAID can be used.
- Prophylactic cytoprotective therapy, e.g. PPI or misoprostol (a synthetic analogue of prostaglandin for all high-risk patients, i.e. over 65 years; those with a peptic ulcer history, particularly with complications, and patients on therapy with corticosteroids or anticoagulants.
- PPIs reduce the risk of endoscopic duodenal and gastric ulcers and are better tolerated than misoprostol, which causes diarrhoea.

Gastrointestinal bleeding

Acute upper gastrointestinal bleeding

- The cardinal features are hematemesis (the vomiting of blood) and melena (the passage of black tarry stools; the black color due to blood altered by passage through the gut).
- Melena can occur with bleeding from any lesion proximal to the right colon (the ligament of Teriz).
- Following a bleed from the upper G tract, unaltered blood can appear per rectum, but the bleeding must be massive and usually accompanied by shock.
- The passage of dark blood and clots without shock is always due to lower GI bleeding.

Etiology

Peptic ulceration is one of the commonest causes of serious and life-threatening gastrointestinal bleeding.

The relative incidence of causes depends on the patient population.

Drugs:

- Aspirin (even 75 mg/day) and other NSAIDs can produce ulcers and erosions. These agents are also responsible for GI hemorrhage from both duodenal and gastric ulcers, particularly in the elderly.
- Corticosteroids in the usual therapeutic doses have no influence on GI hemorrhage.
- Ant-coagulants do not cause acute GI hemorrhage but bleeding from any cause is greater if the patient is anti-coagulated.

Clinical approach to the patient

All cases with a recent (within 48 hours) significant GI bleed should be seen in hospital.

An ABCDE assessment is mandatory to determine whether the patient is unstable or stable.

The ABCDE approach :

Airway (A)

Airway obstruction is an emergency.

Untreated, airway obstruction causes hypoxia to the vital organs, cardiac arrest, and death.

Breathing (B)

During the immediate assessment of breathing, it is vital to diagnose and treat immediately life-threatening conditions (e.g. acute severe asthma, pulmonary oedema, tension pneumothorax, and massive haemothorax).

Circulation (C)

In almost all medical and surgical emergencies, consider hypovolemia to be the primary cause of shock, until proven otherwise.

Unless there are obvious signs of a cardiac cause, give intravenous fluid to any patient with cool peripheries and a fast heart rate.

Disability (D)

Common causes of unconsciousness include profound hypoxia, hypercapnia, cerebral hypoperfusion, or the recent administration of sedatives or analgesic drugs.

Exposure (E)

To examine the patient properly full exposure of the body may be necessary.

The patient's dignity must be respected , also care to minimize heat loss.

In many, no immediate treatment is required as there has been only a small amount of blood loss. Approximately 85% of patients stop bleeding spontaneously within 48 hours.

Emergency management of unstable patient with acute GI bleeding

- **History** and examination. Note co-morbidity
 - ❖ History suggestive of acid – peptic disease
 - ❖ Alcoholic liver diseases / chronic hepatitis / Cirrhosis
 - ❖ History of anticoagulant / anti platelets / NSAIDS / Alcohol binge intake / steroids
 - ❖ History of Coagulation disorder / Blood Dyscrasias
 - ❖ History of Epistaxis or Hemoptysis to rule out the GI source of bleeding
 - ❖ Patients of BURN, Sepsis, Head Trauma may have stress ulcers

- Monitor the pulse and blood pressure half-hourly
- Take blood for hemoglobin, urea, electrolytes, liver biochemistry, coagulation screen, group and cross-matching (2 units initially)
- Establish intravenous access - 2 large bore i.v. cannulae
 - Plasma expanders or 0.9% saline are given until the blood becomes available
- Give **blood transfusion**/colloid if necessary.

Blood volume: The major principle is to rapidly restore the blood volume to normal.

This can be best achieved by transfusion of red cell concentrates via one or more large bore intravenous cannulae; plasma expanders or 0.9% saline are given until the blood becomes available.

The pulse rate and venous pressure are the best guides to adequacy of transfusion.

A central venous pressure line is inserted for patients with organ failure who require blood transfusion, and in those most at risk of developing heart failure.

Indications for blood transfusion are:

- a. SHOCK (pallor, cold nose, systolic BP below 100 mmHg, pulse >100 b.p.m.)
- b. Low Hb in patients with recent or active bleeding

Hemoglobin levels are generally a poor indicator of the need to transfuse because anemia does not develop immediately as hemodilution has not taken place.

However, if the Hb is <10mg/dl and the patient has either bleed recently or is actively bleeding, transfusion is usually necessary.

In most patients, the bleeding stops temporarily, so that further assessment can be made

NB : Transfusion must be monitored to avoid overload leading to heart failure, particularly in the elderly.

- Oxygen therapy should be given by facemask
- Nasogastric tube for gastric wash and monitoring the bleeding the patient should be kept nil by mouth until endoscopy has been performed.
- Urgent endoscopy after stabilizing the patient in shocked patients / liver disease.
- Continue to monitor pulse and BP
- Re-endoscope for continued bleeding/hypovolemia
- Interventional Radiology with embolization of the bleeding vessel also consider surgery

- In addition, stop NSAIDs, aspirin, clopidogrel and warfarin if patients are taking them.
- Stopping antiplatelets can be dangerous and produce thrombosis , an urgent discussion with a cardiologist is needed .
- Patient is best managed by multidisciplinary approach (Medicine , endoscopy , interventional radiology , surgery)
- Patients should be managed in high-dependency beds.
- Patients with large bleeds and clinical signs of shock require urgent resuscitation.

➤ Endoscopy

Endoscopy will usually make a diagnosis and enable therapy to be performed if needed.

Endoscopy should be performed as soon as possible after the patient has been resuscitated.

Endoscopy can detect the cause of the hemorrhage in 80% or more of cases.

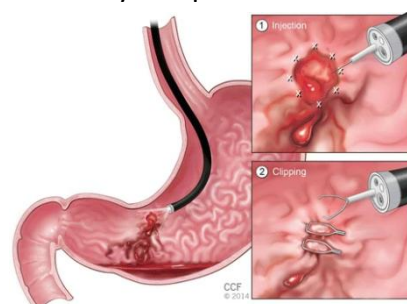
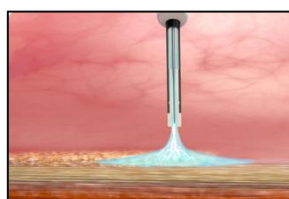
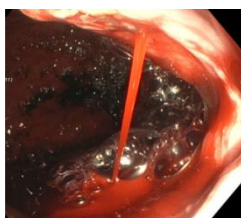
In patients with a peptic ulcer, if the stigmata of a recent bleed are seen (i.e. a spurting vessel, active oozing, fresh or organized blood clot or black spots) the patient is more likely to re-bleed.

At first endoscopy:

Bleeding ulcers and those with stigmata of recent bleeding should be treated with two or three haemostatic methods.

Haemostatic methods:

- ◆ Injection with diluted epinephrine (adrenaline)
 - ◆ thermal coagulation (with heater probe, bipolar probe, laser or argon plasma coagulation)
 - ◆ endoscopic clipping because dual and triple therapy is more effective than monotherapy in reducing rebleeding.
- If varices are present , it should be treated, usually with banding.
 - Antral biopsies should be taken to look for H. pylori.
 - A positive biopsy urease test is valid, but a negative test is not reliable.
 - If the urease test is negative, gastric histology should always be performed.



Injection Therapy

- Injection therapy delivers temporary local tamponade and vasoconstriction
- Use of vasoconstricting agents e.g, epinephrine further reduce blood flow (inject 0.5- to 1.0-mL aliquots of epinephrine (1 : 10,000) via a sclerotherapy needle into four quadrants of the ulcer within 1 to 2 mm of the bleeding site).
- Other agents used: sclerosants like ethanolamine & thrombogenic agents(less efficacious).
- Injection therapy not as efficacious as other modalities of monotherapies
- For actively bleeding ulcers, many experts advocate pre-injection in four quadrants to achieve temporary hemostasis and better visualization

Contact Thermal Therapy

Monopolar Cautery (Cograsper)

Bleeding vessel is compressed and then coagulated.

Low wattage (10-15 watts in duodenum; 15-20 in stomach) is used for a prolonged time (8-12 second pulses) .

End point of treatment is when involved vessel flattens out & there is no bleeding.



Contact Thermal Therapy

Bipolar Cautery

- Thermal modality used most extensively
- For bipolar thermal coagulation, the energy output should be set at 20–25 W with the probe firmly applied on the vessel for several seconds until cavitation or a 'footprint' is obtained
- Provides local tamponade and coaptive coagulation of the underlying vessel



Non-Contact Thermal Therapy

APC is a non-contact method of delivering high-frequency monopolar current through ionized and electrically conductive argon gas

Power/gas flow settings were 50 W and 2 L/min



Mechanical Treatment

- Therapy of choice in obvious arterial bleeding

- Endoscopic hemoclips are widely used.
- Has theoretical advantage over cautery of not causing further tissue damage.
- Re-bleeding rates are reduced with endoclips

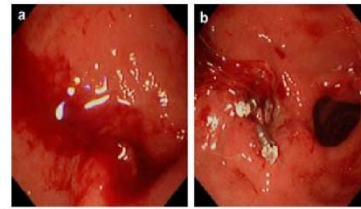


Figure 1. Showing control of bleeding of bleeding

Drug therapy

After diagnosis at endoscopy

- Intravenous omeprazole 80 mg followed by infusion 8 mg/h for 72 hours should be given to all patients with actively bleeding ulcers or ulcers with a visible vessel, as it reduces rebleeding rates and the need for surgery .

Uncontrolled or repeat bleeding:

- Endoscopy should be repeated to assess the bleeding site and to treat, if possible.
- Angiography and embolization of the bleeding vessel by interventional radiology.
- Surgery is necessary if bleeding is persistent and/or uncontrollable and should aim primarily to control the haemorrhage.

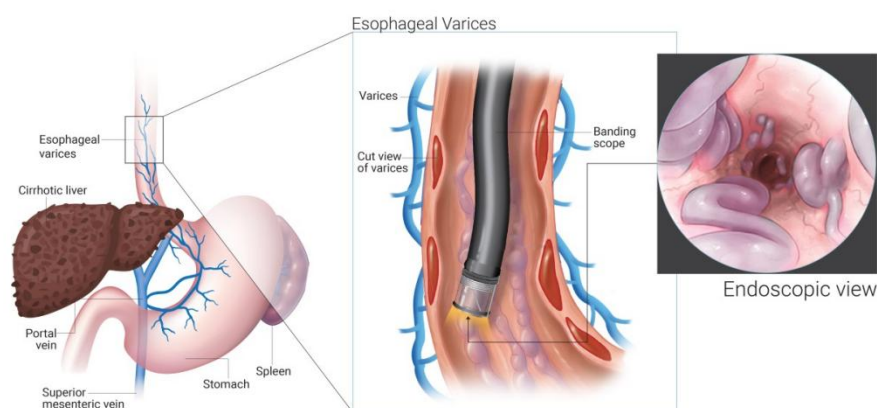
Specific conditions:

Oesophageal varices

About 90 % of patients with liver cirrhosis will develop gastro-esophageal varices over 10 years but only 1/3 of these will bleed.

Esophageal variceal hemorrhage accounts for about 1/3 of deaths in patients with cirrhosis & portal hypertension.

Esophageal Varices



Prevention of the first bleeding:

- Every patient diagnosed with cirrhosis should have an upper GI endoscopy to assess the possibility of presence of esophageal or gastric varices.

- Prophylactic measures:

Non-selective β -blockers:

Carvedilol: a non selective beta and alpha1 receptor blocker is used in dose of 6.25 mg twice daily and is well tolerated .

Propranolol.

The portal inflow is reduced by decrease in cardiac output (β_1) and by blockade of β_2 vasodilator receptor on the splanchnic arteries.

It is as effective as banding.

It prevents bleeding from congestive gastropathy.

Band ligation through the endoscope, is used when there is contraindications or intolerance to β blockers e.g. bronchial asthma, heart block, hypotension.

Management of acute variceal bleeding

1-Resuscitation:

Over blood transfusion may cause rebleeding so Patient should be kept with a hematocrit value around 30.

2- Medical treatment:

PPI IV or H2 blockers.

Vasoconstrictor therapy: for control of emergency bleeding while waiting for endoscopy in combination with endoscopic techniques. They decrease portal inflow by splanchnic arterial constriction.

- Terlipressin.
- Somatostatin and somatostatin analogue (octreotide).

Prophylactic antibiotics: oral or IV.

Measures to prevent encephalopathy.

Vitamin K IV.

Sucralfate to reduce esophageal ulceration after endoscopic therapy.

Patients should have nil by mouth until bleeding has stopped.

Balloon tamponade:

This procedure is rarely used now due to its serious complications as aspiration pneumonia but can be used if endoscopic therapy or vasoconstrictive therapy has failed or not available.

3.Urgent endoscopy: only after resuscitation:

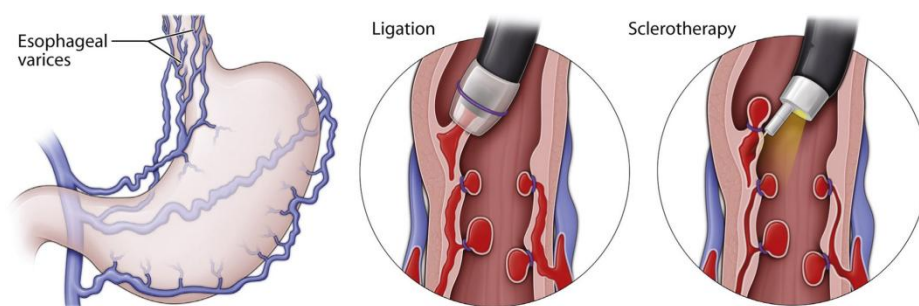
Variceal banding or Injection sclerotherapy.

By band ligation ,the varix is sucked into the end of the scope and a rubber band is dislodged over it trip-wire machine that causes necrosis and fibrosis of the varix .

The varices are injected with a sclerosing agent (like ethanolamine oleate) that produces vessel thrombosis to arrest bleeding.

Gastric varices and big esophageal varices are injected with an adhesive substance (Histoacryl blue) that sticks to the tissues.

Band ligation and Sclerotherapy are the treatment of choice to detect congestive gastropathy.



Management of an acute rebleed

Patients may rebleed within 10 days and in this case endoscopic therapy is repeated.

Trans-jugular intrahepatic portocaval shunt (TIPS):

Is used when bleeding cannot be stopped or early rebleeding occurs after endoscopic therapy within 5 days.

There is an increased risk of portal systemic encephalopathy.

Prevention of recurrent variceal bleeding

Following an episode of variceal bleeding, the risk of recurrence is 60-80% over a 2 years period and this justify measures to prevent rebleeding.

Band ligation: repeated courses every 2 weeks till obliteration of the varices:

Banding is superior to sclerotherapy.

Non- selective beta- blockers.

Surgical porto-systemic shunting shunt : They reduce the risk of rebleeding but produce significant encephalopathy.

Liver transplantation.

Mallory-Weiss tear

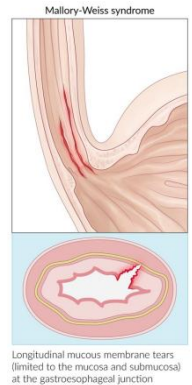
This is a linear mucosal tear occurring at the esophagogastric junction and produced by a sudden increase in intra-abdominal pressure.

It often occurs after a bout of coughing or retching. There may, however, be no antecedent history of retching.

Most bleeds are minor and discharge is usual within 24 hours.

The hemorrhage may be large but most patients stop spontaneously.

Early endoscopy confirms diagnosis and allows therapy.



Chronic peptic ulcer

Eradication of *H. pylori* is started as soon as possible.

A PPI is continued for 4 weeks to ensure ulcer healing.

Eradication of *H. pylori* should always be checked in a patient who has bled and long-term acid suppression given if HP eradication is not achieved.

If bleeding is not controlled the patient should either undergo angiography and embolization or to be referred for surgery.

Gastric carcinoma

Most of these patients do not have large bleeds but surgery is occasionally necessary for uncontrolled or repeat bleeding.

Usually, surgery can be delayed until the patient has been fully evaluated.

Oozing from gastric cancer is very difficult to control endoscopically.

Radiotherapy can occasionally be successful.

Bleeding after percutaneous coronary intervention (PCI)

Many anti-thrombotic medications are in common use after PCI (unfractionated heparin, low molecular weight heparin, , platelet inhibitors e.g. clopidogrel).

All these medications give rise to increased GI bleeding.

A bolus of i.v. PPI could be administered followed by an infusion; platelet infusion can be given to counter the effect of clopidogrel.

Management is difficult as cessation of antiplatelet therapy has a high risk.

Dieulafoy lesion

It is a developmental vascular malformation of the gastrointestinal tract. It is an enlarged submucosal blood vessel that bleeds in the absence of any abnormality, such as ulcers or erosions.

The signs and symptoms are related to blood loss, either due to intermittent or massive gastrointestinal hemorrhage.

To avoid the complications associated with this condition, it must be promptly diagnosed and treated.

Acute lower gastrointestinal bleeding

Massive bleeding from the lower GI tract is rare and usually due to diverticular disease or Ischemic colitis.

Common causes of small bleeds are hemorrhoids and anal fissures.

Management

- The first step is to perform an ABCDE assessment to determine whether the patient is unstable or stable.
- **If patient is unstable**, attempts will be made to stabilize the patient (same principles as upper GI bleeding) also a nasogastric lavage is performed by placing a nasogastric tube and instilling water or normal saline into the stomach.
- The liquid is then aspirated. If blood returns , the upper gastrointestinal source of bleeding is identified.
- On the other hand, if there is no blood, a lower GI bleed is more likely, so you'll need to identify the source.
- If this is the case, angiography, and controlling the bleeding with interventional radiology embolization, or by endoscopic intervention, or in extreme situations surgical intervention
- These procedures are both therapeutic and diagnostic, so besides treating the bleeding, you will be able to find the source and confirm your diagnosis of lower gi bleed.
- **For stable patients**, The first step here is to obtain a focused history and physical exam, including a digital rectal exam, as well as labs like CBC for serial monitoring of hemoglobin and hematocrit, coagulation studies to assess the need to correct a coagulopathy, and blood type and crossmatch.
- In addition, an abdominal and pelvic CT scan with oral contrast to look for pathologies that can cause GI bleed.
- Another important factor is to assess for rectal pain.
- Causes of painless hematochezia include colorectal cancer, Dieulafoy lesion, diverticular bleeding, and angiodysplasia.
- Causes of hematochezia with pain include infectious or inflammatory colitis, or IBD, anorectal causes, like hemorrhoids or anal fissures, and Radiation Telangiectasia or Proctitis.

- Most acute lower GI bleeds start and stop spontaneously.
- The few patients who continue bleeding and are hemodynamically unstable need resuscitation using the same principles as for upper GI bleeding.
- Surgery is rarely required.
- A diagnosis is made using the history, examination including rectal examination and the following investigations as appropriate:
 - ◆ Proctoscopy (e.g. anorectal disease, particularly hemorrhoids)
 - ◆ Flexible sigmoidoscopy or colonoscopy (e.g. inflammatory bowel disease, cancer, ischemic colitis, diverticular disease, angiodysplasia)
 - ◆ Angiography - vascular abnormality (e.g. angiodysplasia). The yield of angiography is low, so it is a test of last resort.
- Isolated episodes of rectal bleeding in the young (<45 years) usually only require rectal examination and flexible sigmoidoscopy because the probability of a significant proximal lesion is very low unless there is a strong family history of colorectal cancer at a young age, Individual lesions are treated as appropriate.

Chronic gastrointestinal bleeding

Patients with chronic bleeding usually present with iron-deficiency anemia.

Chronic blood loss producing iron deficiency anemia in all men, and all women after the menopause, is always due to bleeding from the GI tract.

The primary concern is to exclude cancer, particularly of the stomach or right colon, and celiac disease.

Occult blood tests are unhelpful

Chronic blood loss can occur with any lesion of the GI tract that produces acute bleeding .

However, esophageal varices usually bleed obviously and rarely present as chronic blood loss.

Hookworm is the most common worldwide cause of chronic GI blood loss.

History and examination may indicate the most likely site of the bleeding, but if no clue is available it is usual to investigate both the upper and lower GI tract endoscopically at the same session .

- Upper gastrointestinal endoscopy is usually performed first.
- Duodenal biopsies should always be taken to exclude celiac disease, which is a recognized cause of iron deficiency.
- Colonoscopy follows and any lesion should be biopsied or removed, though it is not safe to assume that colonic polyps are the cause of chronic blood loss.
- Unprepared CT scanning is a reasonable test to look for colon cancer in fragile patients.
- CT colonography can be used as an alternative to colonoscopy.
- If gastroscopy, colonoscopy and duodenal biopsy have not revealed the cause, investigation of the small bowel is necessary.
- Capsule endoscopy is the diagnostic investigation of choice, but currently has no therapeutic ability. Positive diagnostic yield varies from 60% to 85% depending on series. Bleeding lesions can be identified and later treated with balloon-assisted enteroscopy.

Treatment

The cause of the bleeding should be treated, if found.

Oral iron is given to treat anaemia , although intravenous infusions are occasionally required.

Some patients will require maintenance with regular transfusion as a last resort

Prognosis

The mortality from gastrointestinal hemorrhage has not changed from 5-12% over the years, despite many changes in management, mainly because of a demographic shift to more elderly patients with co-morbidity.

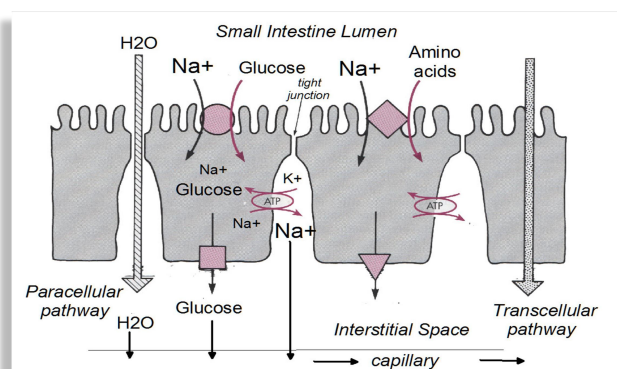
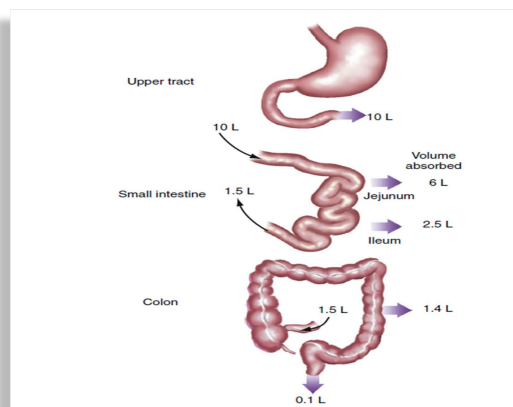
Diarrhea

Basic physiology

- During fasting, the intestine contains very little fluid.
- On a normal diet of three meals a day, **about 9 L** of fluid is delivered to the small intestine.

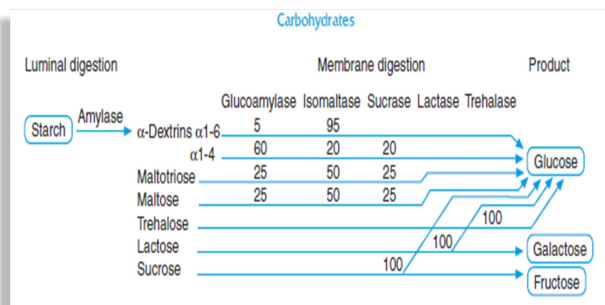
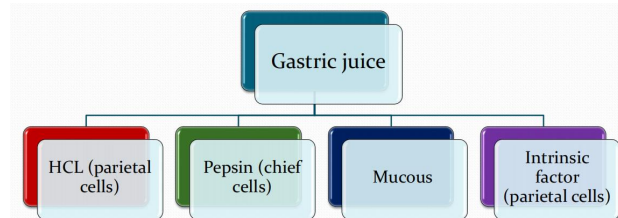
Source	Volume (ml)
☐ Oral intake ☐ Secretions	2,000
1) Salivary	1500
2) Gastric	2500
3) Pancreatic	1500
4) Biliary	500
5) Small intestinal	1000
☐ Total	9000

- **90%** is absorbed in the small intestine.
- **One to two liters** is presented to the colon, and approximately **90% is absorbed**.
- The colon has the capacity to absorb all of the fluid presented to it.
- This results in **100 to 200 mL** of fluid in the stool.
- Feces on the average contain **100 mL of water, 40 mEq/L of sodium (Na⁺), 90 mEq/L of potassium (K⁺), 16 mEq/L of chloride (Cl⁻), and 30 mEq/L of bicarbonate (HCO₃⁻).**
- **Water transport across the intestinal epithelium is passive.**
- It is secondary to osmotic gradients generated by active transport of electrolytes such as Na⁺ and Cl⁻ or other solutes, such as sugars and amino acids.
- **Na⁺** is the major ion actively **absorbed**, and **Cl⁻** is the major ion **secreted**.
- Water is absorbed along with Na⁺.
- **Active Cl⁻ secretion or inhibited Na⁺ absorption** also creates an **osmotic gradient** favorable for the movement of **fluids from blood to lumen** and is the basis of the secretory diarrheas.

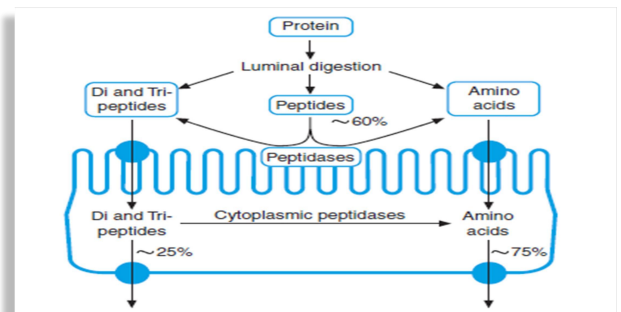


Digestion And Absorption

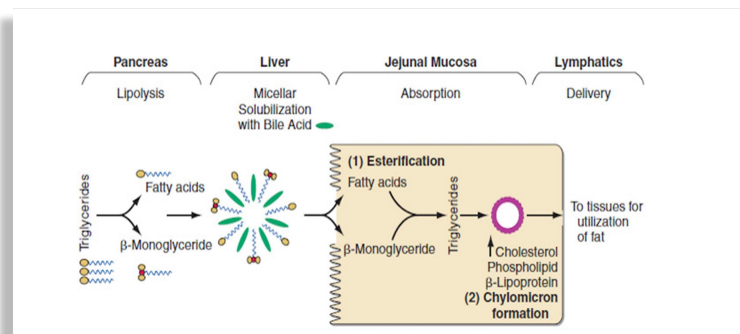
Source	Enzyme
Salivary glands	Amylase
	Lingual lipase
Stomach	Pepsin
Pancreas	Amylase
	Trypsin
	Chymotrypsin
	Carboxypeptidase
	Elastase
	Lipase-colipase
	Phospholipase A ₂
	Cholesterol esterase (nonspecific)
	Enterokinase
	Disaccharidases
	Sucrase
	Maltase
	Trehalase
	α -Dextrinase (isomaltase)
	Peptidases
	Aminooligopeptidase
	Dipeptidase
Intestinal mucosa	



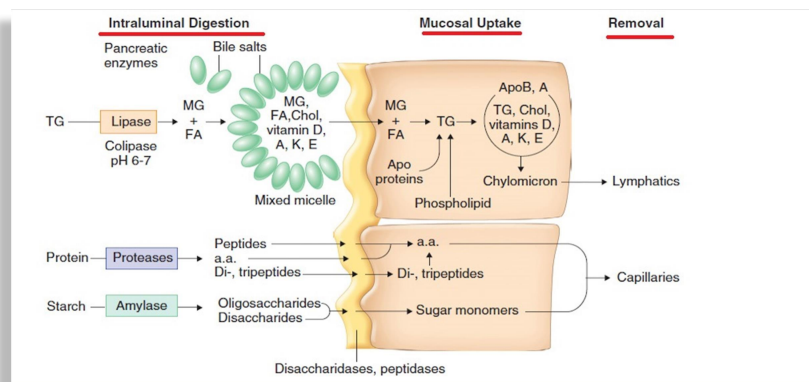
Carbohydrate Digestion And Absorption



Protein Digestion And Absorption



Handling Of Fats By Small Intestine



Overall Digestive And Absorptive Process In Small

Role of bile acids

Bile acids are not present in the diet but are **synthesized in the liver** by a series of enzymatic steps that also include cholesterol catabolism.

Bile acids are either **primary or secondary**: primary bile acids are synthesized in the **liver** from cholesterol, and secondary bile acids are synthesized from primary bile acids in the **intestine** by colonic bacterial enzymes.

The primary functions of bile acids are

- (1) to **promote** bile flow
- (2) to **solubilize** cholesterol and phospholipid in the gallbladder by mixed micelle formation
- (3) **to enhance dietary lipid digestion and absorption** by forming mixed micelles in the proximal small intestine.

Process	Pathophysiologic Defect	Disease Example
Synthesis	Decreased hepatic function	Cirrhosis
Biliary secretion	Altered canalicular function	Primary biliary cirrhosis
Maintenance of conjugated bile acids	Bacterial overgrowth	Jejunal diverticulosis
Reabsorption	Abnormal ileal function	Crohn's disease

Defects Affecting Bile Acids

Diarrhea

Diarrhea is a symptom, not a disease, and therefore may occur in many conditions.

Diarrhea is defined as an increase in the **fluidity, frequency, and volume of daily stool output**.

Normal **stool frequency** ranges from three times a week to three times a day.

Perception as **normal bowel pattern** is related to consistency and frequency.

Consistency is defined as the ratio of fecal water to the water-holding capacity of fecal insoluble solids.

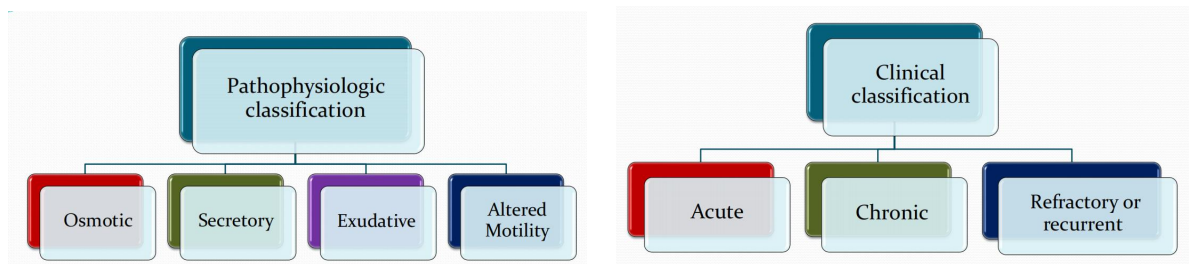
Acute diarrheas are those **lasting less than 4 weeks**.

Approximately 80% of **acute diarrheas** are due to **infections** with viruses, bacteria, and parasites.

Diarrhea **lasting >4 weeks** warrants evaluation to exclude serious underlying pathology.

In contrast to acute diarrhea, **most of the causes of chronic diarrhea are noninfectious**.

Classification of diarrhea



Osmotic diarrhea

There is increased amounts of poorly absorbable, **osmotically active solutes** in the gut lumen.

Osmotic diarrhea is caused by ingestion of a **poorly absorbable solute**, usually a carbohydrate or a divalent ion (e.g., magnesium [Mg²⁺]).

The higher osmolality of the luminal contents **causes water influx into the intestinal lumen** across the duodenal and jejunal epithelium to dilute the solute, in an attempt to make the chyme isotonic

Na⁺ follows the influx of water from the plasma to the gut lumen due to the difference in the Na⁺ concentration gradient.

This **Na⁺ influx causes further influx of water** even after the osmolality of the luminal contents and plasma is identical.

Causes of Osmotic Diarrhea

☐ **Carbohydrate malabsorption:**

I) Disaccharidase deficiency in the small-intestinal mucosa:

- 1) Primary (e.g., idiopathic lactase, sucrase-maltase deficiency)
- 2) Secondary (e.g., after infectious gastroenteritis and with mucosal inflammation)

II) Ingestion of mannitol, sorbitol.

III) Lactulose therapy.

☐ **General malabsorption**

Sprue (idiopathic gluten enteropathy, tropical).

☐ **Ingestion** of sodium sulfate (Glauber's salt), sodium phosphate, magnesium sulfate.

Secretory diarrhea

There is, **increased Cl⁻ and water secretion** with or without inhibition of normal active Na⁺ and water absorption.

Diarrhea of **1 L or more** per day results from secretion of fluid across the intestinal mucosa.

In most cases, a pathophysiologic event causes small intestinal secretion by simultaneously **stimulating active secretion and partial inhibition of intestinal absorption**.

Often, the **intestinal mucosa is intact** and has normal histologic findings.

Clinically, there are features that characterize secretory diarrhea

- 1) Stool volume is usually **large (>1 L per day)**.
- 2) Stools are **watery in consistency**.
- 3) Stools **do not contain pus or blood**.
- 4) Diarrhea typically **continues while the patient fasts** for 24 to 48 hours.

Causes Of Secretory Diarrhea

<u>Enterotoxins</u>	
☐ Vibrio cholerae.	
☐ Staphylococcus aureus.	
<u>Hormonal secretagogues</u>	
☐ Vasoactive intestinal polypeptide (pancreatic cholera syndrome).	
☐ Serotonin (carcinoid).	
<u>Gastric hypersecretion</u>	
☐ Zollinger-Ellison syndrome.	
<u>Laxatives</u>	
☐ Ricinoleic acid (castor oil).	
☐ Dioctyl sodium sulfosuccinate.	
<u>Fatty acids</u>	
☐ Pancreatic insufficiency.	

Exudative diarrhea

There is **mucus, blood, and protein** from sites of active inflammation into the bowel lumen

If the **intestinal mucosa is inflamed and ulcerated**, mucus, blood, and pus **leak into the lumen** and are discharged as stool.

This may also create a **1) increased osmotic load** and If a large surface area of the bowel lumen is involved, **2) absorption of ions, solutes, and water will also be impaired**, and patients may have large volume diarrhea

Inflammation may **3) generate prostaglandins, which stimulate secretion** and may increase bowel motility, thus compounding the diarrhea.

The severity of the diarrhea and the **systemic signs** and symptoms depends on the extent of bowel involvement.

Causes Of Exudative Diarrhea

☐ Idiopathic (e.g., Crohn's disease, ulcerative colitis).

☐ Infectious (e.g., from invasive organisms or cytotoxins: *Shigella, Salmonella, Campylobacter, Yersinia, tuberculosis, amebae, and Clostridium difficile*).

☐ Ischemic.

☐ Vasculitic.

☐ Due to radiation injury.

Altered motility

There is **increased or decreased contact** between luminal contents and mucosal surface.

Increased motility of the small intestine

Results in **decreased contact time** of chyme with absorptive surfaces.

Large amounts of fluid delivered to the colon may **overwhelm its absorptive capacity** and result in diarrhea.

The **reduced contact time in the small intestine** may interfere with absorption of fatty acids and bile salts, allowing them to reach the colon where they induce a **secretory diarrhea**.

decreased motility of the small intestine

May allow **colonization** of the small intestine with colonic-type bacteria.

Clinical classification of diarrhea

Acute diarrhea

Approximately 80% of acute diarrheas are due to **infections** with viruses, bacteria, helminthes, and protozoa.

The remainder are secondary to the ingestion of medications, poorly absorbed sugars (fructose polymers or sorbitol).

Most infectious diarrheas are acquired through **fecal-oral transmission** from water, food, or person-to-person contact.

Patients with infectious diarrhea often complain of nausea, vomiting, and abdominal pain and have watery, malabsorptive, or bloody diarrhea and fever (**dysentery**).

Food poisoning: is produced by a **preformed bacterial toxin that contaminates the food**.

Bacterial replication in the host is not necessary for the development of disease.

Diarrhea resulting from multiplication of organisms in the intestine may be divided into **inflammatory-invasive** versus **noninflammatory-noninvasive** categories.

Epidemiology Of Acute Infectious Diarrhea

Vehicle	Classic Pathogens
Water	Vibrio cholerae, Giardia
Food ☐ Poultry ☐ Seafood and shellfish(Sushi !!!) ☐ Canned vegetables or fruits ☐ Cheese, milk	☐ Salmonella. ☐ Salmonella and Shigella. ☐ Clostridium species ☐ Listeria species
Hospitalization, antibiotics, or chemotherapy	C. difficile
Animal-to-person (pets)	Salmonella, Campylobacter, Cryptosporidium

Causes of Acute Diarrhea According to the mechanism

Causes	Mechanism	Organisms
<u>Viral Infections</u> Small intestine	❖ <u>Mucosal invasion absent</u> ❖ Noninflammatory Watery diarrhea ❖ <u>Fecal leukocyte absent</u>	❖ Rotavirus (children, adults) ❖ Norwalk virus
<u>Bacterial infections</u> ☐ Small intestine ☐ Colon	❖ <u>Mucosal invasion absent</u> ❖ <u>Noninflammatory</u> Watery diarrhea ❖ <u>Fecal leukocytes absent</u> ❖ <u>Mucosal invasion present</u> ❖ <u>Inflammatory</u> Bloody diarrhea ❖ <u>Fecal leukocytes present</u>	❖ Vibrio cholerae ❖ Toxigenic Escherichia coli ❖ Salmonella ❖ Shigella ❖ Invasive E. coli
<u>Parasitic infections</u> ☐ Small intestine ☐ Colon	❖ <u>Mucosal invasion absent</u> ❖ <u>Noninflammatory Watery diarrhea</u> ❖ <u>Fecal leukocyte absent</u> ❖ <u>Mucosal invasion present</u> ❖ <u>Inflammatory</u> Bloody diarrhea ❖ <u>Fecal leukocytes present</u>	❖ Giardia lamblia ❖ Cryptosporidium ❖ Entamoeba histolytica
<u>Food poisoning</u> ☐ Small intestine	❖ <u>Toxin induced</u> ❖ <u>Mucosal invasion absent</u> ❖ Noninflammatory Watery diarrhea ❖ <u>Fecal leukocytes absent</u>	❖ Staphylococcus aureus ❖ Clostridium botulinum

Remarks about treatment of acute diarrhea

Acute diarrhea with fluid and electrolyte depletion is a major cause of **mortality, especially in children** in developing countries of the world.

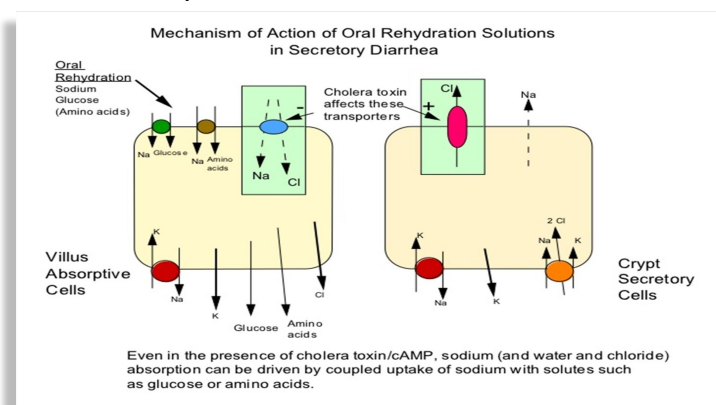
Fluid repletion by intravenous or oral routes can prevent death.

Oral rehydration therapy can be accomplished with a simply prepared oral rehydration solution.

The use of **Opiate derivatives** include paregoric (tincture of opium), diphenoxylate with atropine (Lomotil), and loperamide (Imodium) may help in certain selected cases.

They should not be used in patients with fever, **systemic toxicity, or bloody diarrhea**.

Antimicrobials should be only used when there is **evidence of mucosal invasion**.



Refractory diarrhea

Refractory **chronic diarrhea unresponsive to specific antimicrobial therapy** or standard unspecific medication may present a challenging and serious clinical problem.

Causes

Protracted diarrhea in infants and children

Tropical sprue

Persistent diarrhea in travelers

Post-infectious diarrhea induced irritable bowel syndrome

Protracted infectious diarrhea in adults

In the United States, some organisms typically cause a prolonged course of diarrhea.

Organisms that are difficult to diagnose and are known to cause protracted or prolonged diarrheas include **enteropathogenic (enteroadherent) E. coli** , **Giardia** , amoeba , **Cryptosporidium** , Aeromonas , **Yersinia enterocolitica** , **Clostridium difficile** infection is often difficult to clear, and five or more relapses have been observed .

Clostridium difficile colitis

Originally known as **pseudomembranous colitis**, was first described as an etiology of diarrhea from antibiotic use.

Clostridium difficile infection is triggered by **toxin production from the bacteria**.

Normal bacterial flora is disrupted, the **colon is colonized with the Clostridium difficile bacteria**, and **toxins are released** that cause mucosal damage and inflammation

Diagnosis requires the presence of **1)** diarrhea (at least **3 unformed stools** in 24 h) or **2) radiographic evidence** of ileus or toxic megacolon, in addition to **3)** positive stool testing for **Clostridium difficile toxin** or **4) colonoscopy** or pathologic findings showing pseudo membranes.

Treatment of Clostridium difficile infection involves **discontinuation** of the offending antibiotic.

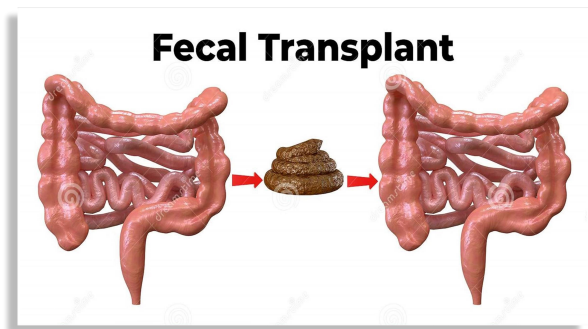
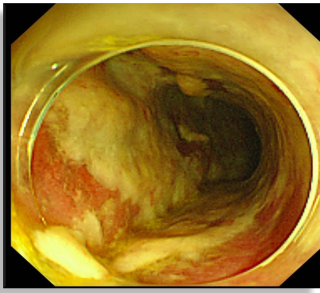
Metronidazole is the drug of choice for mild infection.

Vancomycin should be used for more severe episodes.

Fidaxomicin is a macrocyclic antibiotic which is more active than vancomycin.

The role of **probiotics** in prevention and treatment of Clostridium difficile infection is promising.

Fecal microbiota transplantation has typically been reserved for patients who have refractory Clostridium difficile colitis.



Pseudomembranous Colitis

Post-infectious diarrhea induced irritable bowel syndrome (Brainerd's syndrome)

After documented infectious diarrhea, 25% of patients experience pain, bloating, urgency, a sense of incomplete evacuation, and loose stools for 6 months or longer.

This syndrome of **infectious diarrhea-induced irritable bowel syndrome**, also called **Brainerd's diarrhea** if it is particularly prolonged and accompanied by severe diarrhea.

The pathophysiology is thought to be **unresolved, mild intestinal inflammation**.

In some patients, the syndrome may be the result of **bacterial overgrowth** and may be responsive to non-absorbable antibiotics

In others, it may be the result of **acquired bile acid malabsorption**.

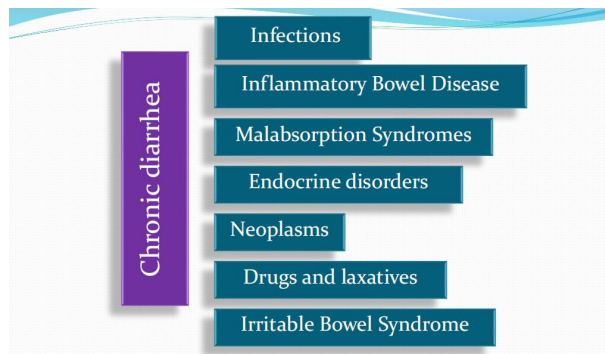
Celiac disease also may manifest after intestinal infections, so patients should be screened for it in this setting

Chronic Diarrhea

Diarrhea lasting **>4 weeks** warrants evaluation to exclude serious underlying pathology.

In contrast to acute diarrhea, most of the causes of chronic diarrhea are **noninfectious**.

Causes



Infections

Giardia lamblia.

Entamoeba histolytica.

Tubercle bacillus.

Clostridium difficile.

Inflammatory bowel disease

Ulcerative colitis and Crohn's disease may result in diarrhea of varying severity, depending on the **extent and degree of bowel involvement**.

Diarrhea in Crohn's disease of the small bowel may be compounded by **concomitant bile salt and fat malabsorption**.

Endocrine disorders

Zollinger-Ellison syndrome.

Hyperthyroidism.

Carcinoid.

Non- β - cell pancreatic tumor.

Malabsorption syndromes

Neoplasms

Drugs and laxatives

Irritable bowel syndrome

TRAVELERS

Bacterial infection (mostly acute)
Protozoal infections (e.g., amebiasis, giardiasis)
Tropical sprue

EPIDEMICS AND OUTBREAKS

Bacterial infection
Epidemic idiopathic secretory diarrhea (e.g., Brainerd diarrhea)
Protozoal infection (e.g., cryptosporidiosis)
Viral infection (e.g., rotavirus, COVID-19*)

PATIENTS WITH DIABETES MELLITUS

Altered motility (increased or decreased)
Associated diseases
Celiac disease
Pancreatic exocrine insufficiency
SIBO
Drug side effects (especially acarbose, metformin)

PATIENTS WITH AIDS

Drug side effects
Lymphoma
Opportunistic infections (e.g., cryptosporidiosis, CMV, HSV, *Mycobacterium avium* complex)

INSTITUTIONALIZED AND HOSPITALIZED PATIENTS

Clostridioides difficile toxin-mediated colitis
Drug side effects
Fecal impaction with overflow diarrhea
Ischemic colitis
Tube feeding

*Coronavirus disease

Malabsorption

In the past, it was believed that most malabsorptive diseases manifested with **diarrhea, steatorrhea, or both**.

It is now recognized that many malabsorptive disorders, such as celiac disease, might have **subtle clinical presentations or mainly extraintestinal manifestations**.

Classically, maldigestion is defined as defective **intraluminal hydrolysis of nutrients**, and malabsorption is defined as defective **mucosal absorption**.

The clinical presentation and complications of maldigestion and malabsorption are **similar**.

Malabsorption can be caused by many diseases of the **small intestine as well as by diseases of the pancreas, liver, biliary tract, and stomach**.

In some of these diseases, malabsorption may be the **presenting feature**; in others, malabsorption may be only a minor clinical problem or detected only as a **laboratory abnormality**.

Malabsorption is caused by **many different diseases**.

Dietary fat is the nutrient **most difficult to absorb**.

Fat is predominantly insoluble in the aqueous milieu of the intestine and critically depends on all phases of digestion, absorption, and delivery for its assimilation.

Steatorrhea (excess fat in the stool) is the hallmark of malabsorption.

Malabsorption Syndromes

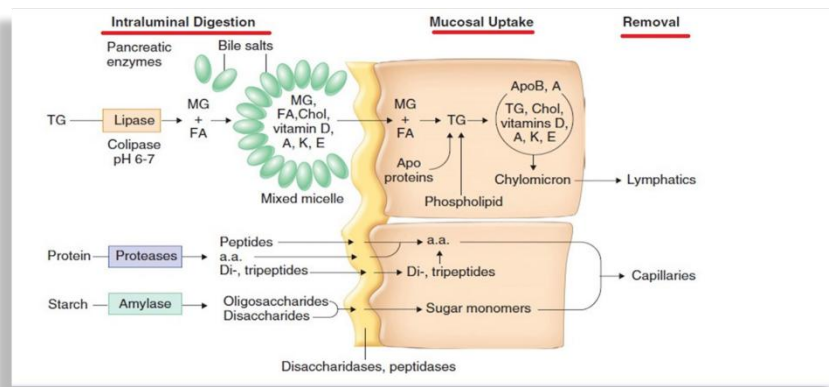
Malabsorption is characterized by **suboptimal absorption** of fats, fat-soluble and other vitamins, proteins, carbohydrates, electrolytes and minerals, and water.

At the most basic level, it is the result of disturbance of at least one of these normal digestive functions:

1) Intraluminal digestion: The process begins in the mouth with saliva, receives a major boost from **gastric peptic digestion**, and continues in the small intestine, assisted by **pancreatic enzyme secretion** and the **emulsive action of bile**.

2) Terminal digestion: which involves the hydrolysis of carbohydrates and peptides by **disaccharidases and peptidases**, respectively, in the **brush border** of the **small intestinal mucosa**.

3) Transepithelial transport: in which nutrients, fluid, and electrolytes are transported across the epithelium of the small intestine for **delivery to the intestinal vasculature or to the intestinal lymphatic system**.



Overall digestive and absorptive process in small intestine

The Major Malabsorption Syndromes

Disorders of absorption constitute a broad spectrum of conditions with **multiple etiologies and varied clinical manifestations**.

Most, but not all, malabsorption syndromes are associated with **steatorrhea, an increase in stool fat excretion of >6% of dietary fat intake**.

Some malabsorption disorders are not associated with steatorrhea.

Disorders of absorption (malabsorption) must be included in the **differential diagnosis of chronic diarrhea**.

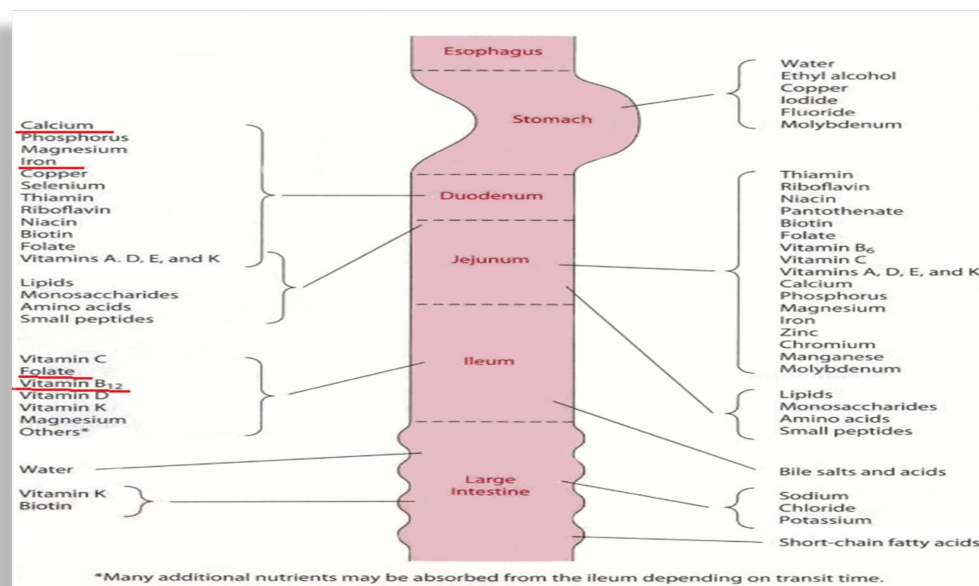
Most patients will indicate that they have diarrhea, not that they have fat malabsorption.

Mechanism of malabsorption according to the condition

MECHANISM OF MALABSORPTION	CONDITIONS
Impaired mixing	➤ Partial/total gastrectomy ➤ Gastric bypass surgery
Impaired lipolysis	➤ Chronic pancreatitis ➤ Pancreatic cancer
Impaired micelle formation (related to bile acids)	➤ Severe chronic liver disease ➤ Cholestatic liver disease ➤ Bacterial overgrowth ➤ Crohn disease ➤ Ileal resection
Impaired mucosal absorption	➤ Giardiasis ➤ Celiac disease ➤ Tropical sprue ➤ Radiation enteritis ➤ Whipple disease ➤ Lymphoma
Impaired nutrient delivery	➤ Lymphoma ➤ Tuberculosis ➤ Constrictive pericarditis ➤ Severe congestive heart failure

Clinical Consequences Of Malabsorption Of Nutrients, Water, And Electrolytes

Nutrient Malabsorbed	Clinical Manifestation
Protein	Wasting, edema
Carbohydrate and fat	Diarrhea, abdominal cramps and bloating, weight loss/growth retardation
Fluid and electrolytes	Diarrhea, dehydration
Iron	Anemia, cheilosis, angular stomatitis
Calcium/vitamin D	Bone pain, fractures, tetany
Magnesium	Paresthesia, tetany
Vitamin B12/folate	Anemia, glossitis, cheilosis, paresthesias, ataxia (vitamin B12 only)
Vitamin A	Night blindness, xerophthalmia, hyperkeratosis, diarrhea
Vitamin K	Ecchymoses
Riboflavin	Angular stomatitis, cheilosis
Selenium	Cardiomyopathy
Essential fatty acids	Dermatitis



Specific disease examples

Lactase Deficiency (lactose intolerance)

Acquired lactase deficiency is the **most common cause of selective carbohydrate malabsorption**.

The prevalence of lactase deficiency is highest (85 to 100%) in persons of **Asian, African, and Native-American descent**.

Adults with lactase deficiency typically complain of gas, bloating, and diarrhea after the **ingestion of milk or dairy products but do not lose weight**.

Unabsorbed lactose is osmotically active, **drawing water followed by ions into the intestinal lumen**.

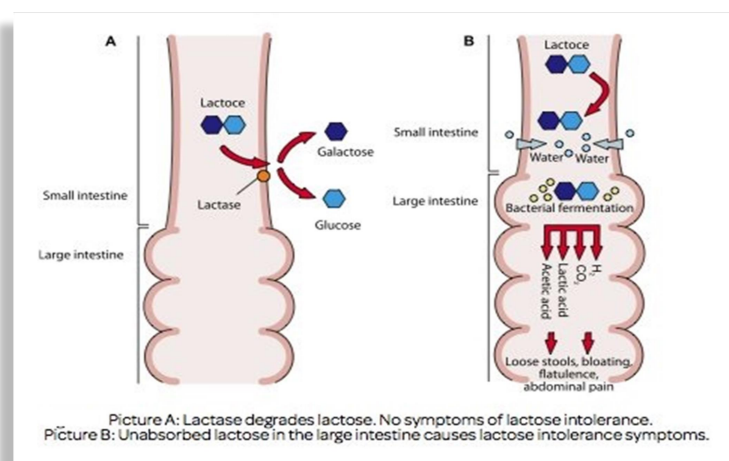
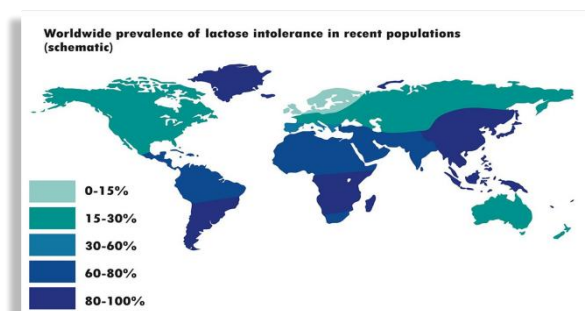
On reaching the colon, bacteria metabolize lactose to **short-chain fatty acids, carbon dioxide, and hydrogen gas**.

Short-chain fatty acids are transported with sodium into colonic epithelial cells, facilitating the reabsorption of fluid in the colon.

If the colonic capacity for the **reabsorption of short-chain fatty acids is exceeded, an osmotic diarrhea results**.

The diagnosis of acquired lactase deficiency can be made by **1) empirical treatment with a lactose-free diet**, which results in resolution of symptoms;**2) by the hydrogen breath test** after oral administration of lactose.

Many intestinal diseases cause **secondary reversible lactase deficiency**, including viral gastroenteritis, celiac disease, giardiasis, and bacterial overgrowth.



Intestinal Bacterial Overgrowth

In health, only small numbers of lactobacilli, enterococci, gram-positive aerobes, or facultative anaerobes **can be cultured from the upper small bowel lumen**.

Motility And Acid are the most important factors in keeping the number of bacteria in the upper small bowel low.

Any condition that produces local stasis or re-circulation of colonic luminal contents allows development of a predominantly **colonic flora (coliforms and anaerobes, such as Bacteroides and Clostridium) in the small intestine**.

This will cause **impaired micelle formation** by releasing **cholyamidases**, which **deconjugate bile salts**.

As a result, the concentration of bile salts decreases in the intestinal lumen causing **malabsorption of fats and fat-soluble vitamins**.

Vitamin B12 deficiency and carbohydrate malabsorption also can occur with generalized bacterial overgrowth.

Anaerobic bacteria **ingest vitamin B12 and release proteases that degrade brush-border disaccharidases**.

Individuals with bacterial overgrowth usually have **low serum vitamin B12 levels but normal or high folate levels**.

This helps distinguish bacterial overgrowth from **tropical sprue**.

Some Causes Of Bacterial Overgrowth

STRUCTURAL	
<u>Surgical</u>	☐ Ileocecal valve resection
<u>Anatomic</u>	☐ Duodenal and jejunal diverticula ☐ Gastrojejunocolic fistulas
MOTOR	
	☐ Scleroderma ☐ Diabetes mellitus
HYPOCHLORHYDRIA	
	☐ Atrophic gastritis ☐ Proton-pump inhibitors
MISCELLANEOUS	
	☐ Cirrhosis ☐ Chronic renal failure

Celiac Disease

Celiac disease, also called celiac sprue, nontropical sprue, and gluten-sensitive enteropathy, is an **inflammatory condition of the small intestine** precipitated by the ingestion of wheat, rye, and barley in individuals with certain genetic predispositions.

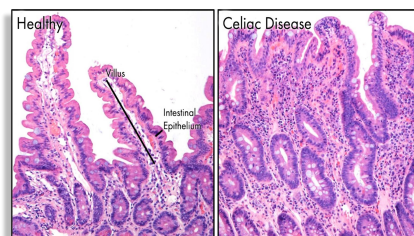
It is the prototype of a **noninfectious cause of malabsorption** resulting from a reduction in small intestinal absorptive surface area

The basic disorder in celiac disease is **sensitivity to gluten**, the component of wheat and related grains (oat, barley, and rye) that **contains the water insoluble protein gliadin**.

The small intestinal mucosa, when exposed to gluten, accumulates **large numbers of B cells and plasma cells sensitized to gliadin**; accumulation of lymphocytes in gastric and colonic mucosa also may occur.

Total flattening of mucosal villi (and hence loss of surface area) is the outcome, affecting the **proximal more than the distal small intestine**.

Patients with celiac disease have **increased levels of serum antibodies** to a variety of antigens, including IgA antiendomysial, anti gliadin and anti tissue transglutaminase antibodies (**TTG Ab**).



The **age of presentation** with symptomatic diarrhea and malnutrition varies from infancy to mid-adulthood.

About half of adults with celiac disease in the present with **anemia or osteoporosis without diarrhea** or other gastrointestinal symptoms.

These individuals most likely have **proximal disease** that impairs iron, folate, and calcium absorption but an adequate surface area in the remaining intestine for absorption of other nutrients.

Other **Extraintestinal Manifestations Of Celiac Disease** include

- Rash (dermatitis herpetiformis)
- Neurologic disorders (peripheral neuropathy, ataxia, epilepsy)
- Psychiatric disorders (depression, paranoia)
- Reproductive disorders (infertility, spontaneous abortion)
- Short stature
- Dental enamel hypoplasia
- Chronic hepatitis

● Cardiomyopathy.

Manifestation	Probable Cause(s)
CUTANEOUS	
Echymoses and petechiae	Vitamin K deficiency; rarely, thrombocytopenia
Edema	Hypoproteinemia
Dermatitis herpetiformis	Epidermal (type 3) tTG autoimmunity
Follicular hyperkeratosis and dermatitis	Vitamin A malabsorption, vitamin B complex malabsorption
ENDOCRINOLOGIC	
Short stature, delayed puberty	Malnutrition, hypothalamic-pituitary dysfunction
Amenorrhea, infertility, impotence	Malnutrition, hypothalamic-pituitary dysfunction, immune dysfunction
Secondary hyperparathyroidism	Calcium and/or vitamin D malabsorption with hypocalcemia
HEMATOLOGIC	
Anemia	Iron, folate, or vitamin B ₁₂ deficiency
Hemorrhage	Vitamin K deficiency; rarely, thrombocytopenia due to folate deficiency
Thrombocytosis, Howell-Jolly bodies	Hyposplenism
HEPATIC	
Elevated liver biochemical test levels	Lymphocytic hepatitis
Autoimmune hepatitis	Autoimmunity
MUSCULAR	
Atrophy	Malnutrition due to malabsorption
Weakness	Generalized muscle atrophy, hypokalemia
NEUROLOGIC	
Peripheral neuropathy	Deficiencies of vitamin B ₁₂ and thiamine; immune-based neurologic dysfunction
Ataxia	Cerebellar and posterior column damage
Demyelinating CNS lesions	Immune-based neurologic dysfunction
Seizures	Unknown
SKELETAL	
Osteopenia, osteomalacia, and osteoporosis	Malabsorption of calcium and vitamin D, secondary hyperparathyroidism, chronic inflammation

ASSOCIATED CONDITIONS

Addison disease
 Autoimmune hemolytic anemia
 Autoimmune liver diseases
 Bird-fancier's lung
 Cavitory lung disease
 Cystic fibrosis
 Dermatitis herpetiformis
 Diabetes mellitus type 1
 Down syndrome
 Epilepsy with cerebral calcification
 Fibrosing alveolitis
 Hypothyroidism or hyperthyroidism
 Idiopathic pulmonary hemosiderosis
 Immune thrombocytopenic purpura
 Immunoglobulin (Ig)A deficiency
 Iridocyclitis or choroiditis
 Macroamylasemia
 Microscopic colitis
 Recurrent pericarditis
 Sarcoidosis
 Sjögren syndrome

INCREASED MORBIDITY AFTER CELIAC DISEASE

DIAGNOSIS

Cardiovascular: ischemic heart disease, cerebrovascular disease, atrial fibrillation, dilated cardiomyopathy
 Malignancy: non-Hodgkin lymphoma, small intestinal adenocarcinoma
 Respiratory: chronic obstructive pulmonary disease, asthma, tuberculosis, influenza

Disease association with celiac disease

Individuals with **significant** mucosal involvement present with **watery diarrhea**, **weight loss** or growth retardation, and the clinical manifestations of **vitamin and mineral deficiencies**.

Diarrhea is caused by many mechanisms, including

- 1) **Decreased surface area** for water and electrolyte absorption.
- 2) **The osmotic effect of unabsorbed luminal nutrients**
- 3) **Increased surface area for chloride secretion** (crypt hyperplasia)
- 4) Stimulation of intestinal fluid secretion by **inflammatory mediators** and unabsorbed fatty acids.

Recently The perception that **gluten** causes gastrointestinal and extra gastrointestinal symptoms in **patients who do not have coeliac disease (CD)**, and are not allergic to wheat, is an increasing clinical problem.

This has been called **Non-coeliac Gluten Sensitivity (NCGS)**.

It is purely a clinical diagnosis as, in contrast to CD, it has no biomarker or characteristic intestinal biopsy findings.

Tropical Sprue And Whipple Disease

Are two disorders that exemplify malabsorption syndromes arising from **intestinal infection**.

Tropical sprue resembles celiac disease in symptomatology but occurs almost exclusively in persons **living in or visiting the tropics**.

Small intestinal changes vary from near normal to a severe diffuse enteritis with villus flattening.

In contrast to celiac disease, **injury is seen at all levels of the small intestine**.

It is caused by overgrowth of predominantly **coliform bacteria**.

Individuals with tropical sprue classically present with **diarrhea and megaloblastic anemia** secondary to vitamin B12 and folate deficiency, but some have anemia only.

Intestinal biopsy characteristically shows subtotal and **patchy villous atrophy in the proximal and distal small intestine**.

Treatment is a **prolonged course of tetracycline (250 mg PO four times daily) or doxycycline (100 mg PO two times daily), folic acid (5 mg/day PO)**, and, with coexistent deficiency, **vitamin B12 injections (1000 µg weekly)**.

Whipple disease is a rare, systemic infection that may involve any organ of the body but principally affects the **Intestine, Central Nervous System, And Joints**.

The hallmark of Whipple disease is a small intestinal mucosa laden with distended **periodic acid-Schiff (PAS)-positive macrophages** in the lamina propria.

The organism responsible for causing Whipple disease is a gram-positive actinomycete, **Tropheryma whippelii**.

Approximately one third of patients have **cardiac involvement**, most commonly culture negative endocarditis.

Occasionally, individuals present with **ocular or neurologic disease without gastrointestinal symptoms**.

Men are affected more commonly than women, particularly white men.

Treatment is with a **prolonged course of broad-spectrum antibiotics** (e.g., ceftriaxone, 2 g/day intravenously (IV) or meropenem 1 g IV three times daily; then **trimethoprim 160 mg and sulfamethoxazole 800 mg PO two times daily for 1 year**).

Diagnostic tests for malabsorption

1) General tests of absorption

Quantitative stool fat test Gold standard test of fat malabsorption.

Requires ingestion of a high-fat diet (70- 100 g) for 2 days before and during the collection.

Stool is collected for 2-3 days. Normally, <7 g/24 hr is excreted on a high-fat diet

Qualitative stool fat test Sudan stain of a stool sample for fat.

D-Xylose test A test of small intestinal mucosal absorption, used to distinguish mucosal malabsorption from malabsorption due to pancreatic insufficiency.

Hydrogen breath test

Most useful in the diagnosis of lactase deficiency.

An oral dose of lactose (25 g) is administered after measurement of basal breath H₂ levels.

The sole source of H₂ in the mammal is bacterial fermentation; unabsorbed lactose makes its way to colonic bacteria, resulting in excess breath H₂

2) Tests for Pancreatic Function

Secretin stimulation test

The gold standard test of pancreatic function

Fecal elastase-1 test

Stool test for pancreatic function

3) Tests for Bacterial Overgrowth

Quantitative culture of small intestinal aspirate

Gold standard test for bacterial overgrowth

Hydrogen breath test

The 50-g glucose breath test has a sensitivity of 90% for growth of 10⁵ colonic-type bacteria in the small intestine

¹⁴C-D-xylose breath test

This test uses 1 g of carbon ¹⁴-labeled D-xylose.

It has a sensitivity and specificity >90%

4) Tests for Mucosal Disease

Small-bowel biopsy: Distal duodenal biopsy specimens are usually adequate for diagnosis.

Permeability studies: These tests of mucosal integrity are gaining favor as screening tests for small intestinal disease and for monitoring response to treatment

5) Test of Ileal Function

SeHCAT test

This is a test of bile acid absorption.

Seven days after ingestion of **radiolabeled synthetic selenium–homocholic acid conjugated with taurine (SeHCAT)**, whole body retention is measured by a gamma-counting device.

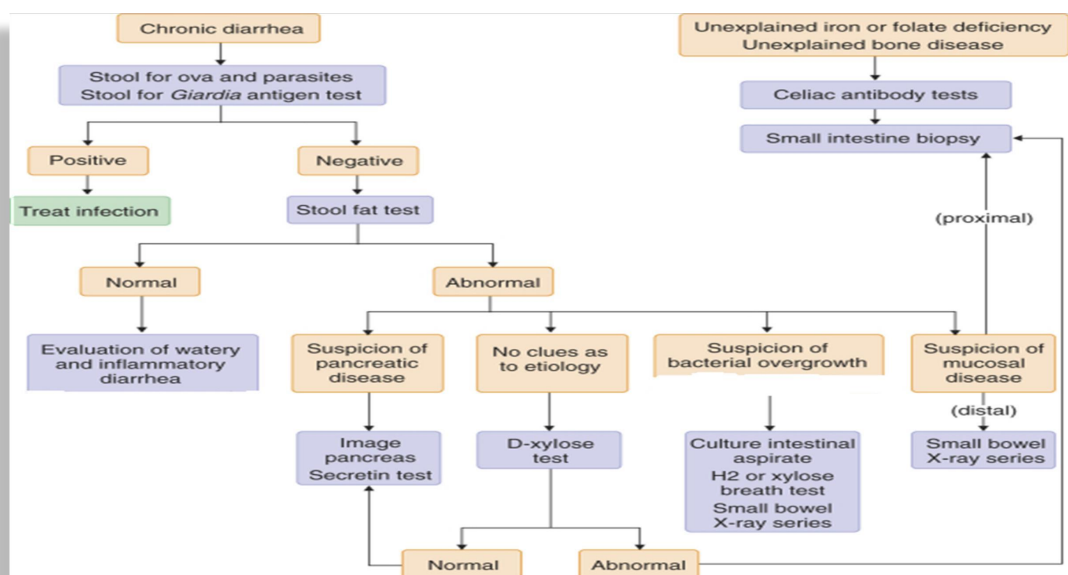
The result is expressed as a fraction of baseline ingestion

Schilling test: No longer used

GENERAL TESTS OF ABSORPTION	
Quantitative stool fat test	Gold standard test of fat malabsorption, with which all other tests are compared. Requires ingestion of a high-fat diet (100 g) for 2 days before and during the collection. Stool is collected for 3 days. Normally, <7 g/24 hr is excreted on a high-fat diet. Borderline abnormalities of 8-14 g/24 hr may be seen in secretory or osmotic diarrheas that are not caused by malabsorption. There are false-negative findings if fat intake is inadequate. False-positive results can occur if the nonabsorbable fat olestra is ingested or mineral oil laxatives or rectal suppositories (e.g., cocoa butter) are given to the patient before stool collection.
Qualitative stool fat test	Sudan stain of a stool sample for fat. Many fat droplets per medium-power field (40×) constitute a positive test result. The nuclear magnetic resonance method determines the percentage of fat in the stool (normal, <20%). The test depends on an adequate fat intake (100 g/day). There is high sensitivity (90%) and specificity (90%) with fat malabsorption of >10 g/24 hr. Sensitivity drops with stool fat in the range of 6-10 g/24 hr.
Hydrogen breath test	Most useful in the diagnosis of lactase deficiency. An oral dose of lactose (1 g/kg body weight) is administered after measurement of basal breath H ₂ levels. The sole source of H ₂ in the mammal is bacterial fermentation; unabsorbed lactose makes its way to colonic bacteria, resulting in excess breath H ₂ . A late peak (within 3-6 hr) of > 20 ppm of exhaled H ₂ after lactose ingestion suggests lactase malabsorption. Absorption of other carbohydrates (e.g., sucrose, glucose, fructose) also can be tested.
SPECIFIC TESTS FOR MALABSORPTION	
Tests for Pancreatic Function	
Secretin stimulation test	The gold standard test of pancreatic function. Requires duodenal test intubation with a double-lumen tube and collection of pancreatic juice in response to intravenous secretin. Allows measurement of bicarbonate (HCO ₃ ⁻) and pancreatic enzymes. A sensitive test of pancreatic function, but labor intensive and invasive.
Fecal elastase-1 test	Stool test for pancreatic function. Equal sensitivity to the secretin stimulation test for the diagnosis of moderate-to-severe pancreatic insufficiency. More specific than the fecal chymotrypsin test. Unreliable with mild insufficiency. False-positive results occur with increased stool volume and intestinal mucosal diseases.
Tests for Bacterial Overgrowth	
Quantitative culture of small intestinal aspirate	Gold standard test for bacterial overgrowth. Greater than 10 ⁵ colony-forming units (CFU)/mL in the jejunum suggests bacterial overgrowth. Requires special anaerobic sample collection, rapid anaerobic and aerobic plating, and care to avoid oropharyngeal contamination. False-negative results occur with focal jejunal diverticula and when overgrowth is distal to the site aspirated.
Hydrogen breath test	The 50-g glucose breath test has a sensitivity of 90% for growth of 10 ⁵ colonic-type bacteria in the small intestine. If bacterial overgrowth is present, increased H ₂ is excreted in the breath. A hydrogen level (within 2 hr) of > 20 ppm suggests bacterial overgrowth. False-negative results occur with non-hydrogen-producing organisms. Concomitant measurement of breath methane improves test sensitivity.
Tests for Mucosal Disease	
Small bowel biopsy	Obtained for a specific diagnosis when there is a high index of suspicion for small intestinal disease. Several biopsy specimens (4-5) must be obtained to maximize the diagnostic yield. Distal duodenal biopsy specimens are usually adequate for diagnosis, but occasionally enteroscopy with jejunal biopsy specimens is necessary. Small intestinal biopsy provides a specific diagnosis in some diseases (e.g., intestinal infection, Whipple disease, abetalipoproteinemia, agammaglobulinemia, lymphangiectasia, lymphoma, amyloidosis). In other conditions, such as celiac disease and tropical sprue, the biopsy specimens show characteristic findings, but the diagnosis is made on improvement after treatment.
Tests of Ileal Function	
Schilling test	A test of vitamin B ₁₂ absorption ()
⁷⁵ SeHCAT test	This is a test of bile acid absorption. Seven days after ingestion of radiolabeled synthetic selenium–homocholic acid conjugated with taurine (⁷⁵ SeHCAT), whole body retention is measured by a gamma-counting device. The result is expressed as a fraction of baseline ingestion. Retention values of <10% are abnormal and indicate bile acid malabsorption with a sensitivity of 80-90% and specificity of 70-100%. The radiation dose is equivalent to that of a plain chest x-ray. Liver disease and bacterial overgrowth may give false results. Not approved for use in the United States.

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Acid steatocrit	Reliable screening test for fat malabsorption that is inexpensive and easy to perform. Centrifugation of acidified stool in a hematocrit capillary yields solid, liquid, and fatty layers. Results are expressed as volumetric percentages (lipid phase on solid phase); normal, <10%. High sensitivity (100%) and specificity (95%) compared with the 72-hr stool quantitative fat test. Depends on adequate fat intake (100 g/day).
D-Xylose test	A test of small intestinal mucosal absorption, used to distinguish mucosal malabsorption from malabsorption due to pancreatic insufficiency. An oral dose of D-xylose (25 g/500 mL water) is administered, and D-xylose excretion is measured in a 5-hr urine collection. Normally, >4 g of D-xylose is excreted in the urine over 5 hr. The test also may be positive in bacterial overgrowth owing to metabolism of D-xylose by bacteria in the intestinal lumen. False-positive test results occur with renal failure, ascites, and an incomplete urine collection. Blood levels at 1 and 3 hr improve sensitivity. May be normal with mild or limited mucosal disease.
Hydrogen breath test	Most useful in the diagnosis of lactase deficiency. An oral dose of lactose (1 g/kg body weight) is administered after measurement of basal breath H ₂ levels. The sole source of H ₂ in the mammal is bacterial fermentation; unabsorbed lactose makes its way to colonic bacteria, resulting in excess breath H ₂ . A late peak (within 3–6 hr) of >20 ppm of exhaled H ₂ after lactose ingestion suggests lactose malabsorption. Absorption of other carbohydrates (e.g., sucrose, glucose, fructose) also can be tested.
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TESTS FOR BACTERIAL OVERGROWTH	
Quantitative culture of small intestinal aspirate	Gold standard test for bacterial overgrowth. Greater than 10 ⁶ colony-forming units (CFU)/mL in the jejunum suggests bacterial overgrowth. Requires special anaerobic sample collection, rapid anaerobic and aerobic plating, and care to avoid oropharyngeal contamination. False-negative results occur with focal jejunal diverticula and when overgrowth is distal to the site aspirated.
Hydrogen breath test	The 50-g glucose breath test has a sensitivity of 90% for growth of 10 ⁶ colonic-type bacteria in the small intestine. If bacterial overgrowth is present, increased H ₂ is excreted in the breath. An early peak (within 2 hr) of >20 ppm exhaled H ₂ suggests bacterial overgrowth. False-negative results occur with non-hydrogen-producing organisms.
14C-D-xylose breath test	This test uses 1 g of carbon 14-labeled D-xylose. It has a sensitivity and specificity >90% for growth of 10 ⁶ colonic-type bacteria in the small intestine. Bacteria metabolize D-xylose with release of 14CO ₂ , which is absorbed and exhaled. Non-degraded D-xylose is absorbed in the small bowel and does not reach the colon, yielding a greater specificity than the H ₂ breath test. A nonradioactive 13C-D-xylose breath test is suitable for children and pregnant women.

Algorithm To Chronic Diarrhea And Malabsorption



Constipation

Constipation is a **common** gastrointestinal condition characterized by **infrequent bowel movements** and **difficulty passing stool**.

It happens when the intestines **don't move waste** to the rectum properly, and pelvic muscles and **anal sphincter** don't coordinate well to expel stool.

Based on the cause, constipation can be classified as

- **Primary**, also known as idiopathic or functional constipation.
- **Secondary**, which occurs as a side effect of some medications or due to another medical condition, such as malignancy.

A focused **history** and **physical examination** including digital rectal examinations are mandatory.

The patient will report a history of **infrequent bowel movements**, usually fewer than three stools per week, as well as **straining**, **hard stools**, and a feeling of **incomplete evacuation**.

Additionally, history might reveal **abdominal discomfort or bloating**, and sometimes the patient might report the use of **manual maneuvers** to defecate.

Next, in patients with **acute onset constipation**, always ask about the ability to pass gas, as failure to pass gas can signal bowel obstruction.

Other symptoms that point to obstruction include **obstipation**, which refers to severe or complete constipation with practically no stool passage and absence of flatus, and can be accompanied by abdominal pain, nausea, and vomiting.

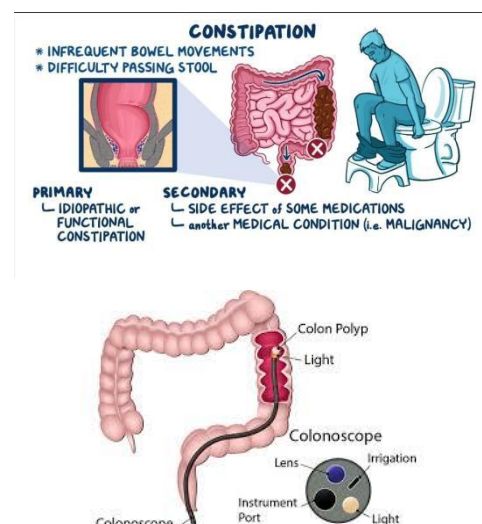
By physical examination findings, which typically include **fecal impaction** and a palpable stool ball.

Additionally, some patients may have **anal fissures**, **hemorrhoids**, or **functional problems** like **pelvic floor Dys-synergia**, this is when the anal sphincter **fails to relax** or the perineum does not descend when the patient bears down during evacuation.

Based on these history and physical exam findings, it's important to look for **red flag features** that could indicate underlying serious conditions like malignancy.

Red flag features:

- Hematochezia
- Unintentional weight loss
- A family history of colorectal cancer
- Acute onset of constipation in an older adult
- Change in stool caliber
- Anemia
- The presence of a rectal mass.



If any of these red flag features are present, we should consider colorectal cancer and proceed with an **endoscopy**, such as colonoscopy, with a biopsy to look for obstructive lesions.

For patients with no red flag features, to assess for **secondary causes** of constipation.

Secondary causes:

- Medication side effects, commonly seen in patients taking opioid medications or iron supplements.
- Endocrine or metabolic disorders like hypothyroidism, hypokalemia, or hypercalcemia.
- Neurogenic conditions, such as spinal cord injury or multiple sclerosis.

In primary constipation a trial of empiric therapy, which includes increasing dietary fiber or starting a **fiber** supplement, advising sedentary patients to increase their **physical activity**, and starting an over-the-counter **laxative**, such as an osmotic agent or a stimulant laxative.

If there is an adequate response to empiric therapy no further workup is needed.

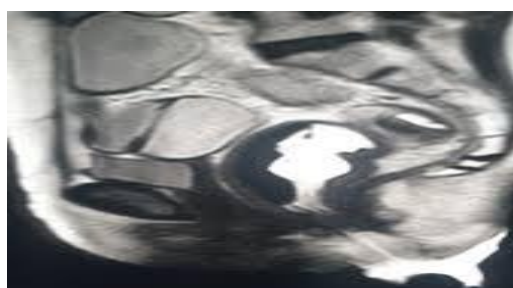


On the other hand, if they have an inadequate response to empiric therapy, **additional testing** will be mandatory as **anorectal manometry** and a **balloon expulsion test** or **defecography**.

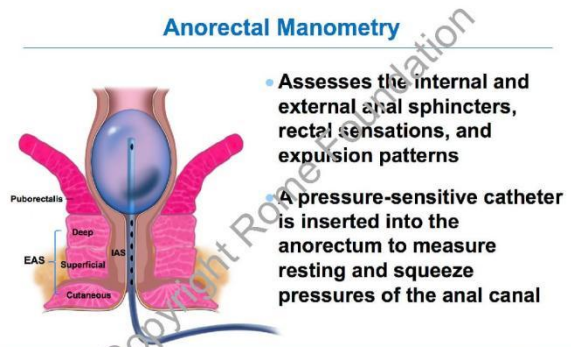
These tests measure pressures inside the rectum and anus and the **ability of the pelvic muscles to expel stool** from the rectum.

These test are abnormal in defecatory disorder.

In this condition constipation arises from **difficulty in evacuating** stool from the rectum.



MRI Defecography



However, if the anorectal manometry and balloon expulsion test are normal, a **colonic transit** study will be recommended.

During colonic transit studies, patients swallow radioopaque markers, which are then tracked through the gastrointestinal tract using X-rays.

If the markers are not passed within the expected time frame, then the test is abnormal, so you can diagnose **slow transit constipation**.

Colonic Transit Study (Sitzmarker Technique)



On other side, if the colonic transit study results are normal, diagnose **normal transit constipation**, which is also referred to as a disorder of the gut-brain axis.

There are different types of normal transit constipation.

One type is irritable bowel syndrome, which might be marked by alternating episodes of diarrhea and constipation, or dominated by either symptom.

Functional Bowel disorders and Irritable Bowel Syndrome

Irritable bowel syndrome

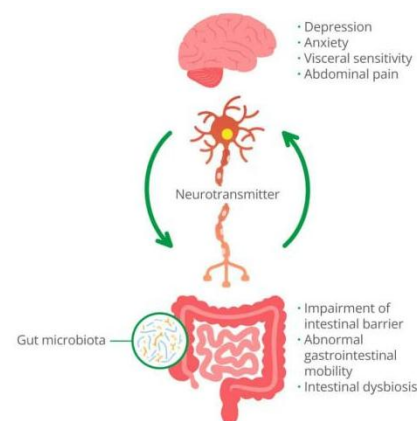
Functional dyspepsia

Functional chest pain of presumed esophageal origin

Functional heartburn

characterized by chronic, recurrent symptoms of pain and discomfort referred to the lower abdomen, the epigastrium and upper abdomen, and the retrosternum, respectively.

They belong to the family of functional gastrointestinal (GI) disorders (also referred to as disorders of brain-gut interactions) that comprise a wide spectrum of often overlapping chronic GI disorders that are common in both the adult and pediatric populations.



Irritable bowel syndrome or IBS

It is a **chronic** bowel condition characterized by recurrent abdominal **pain** associated with abnormal bowel movements.

The cause is **unknown** but could be related to **changes in the normal gut microbiota**, **autonomic dysfunction**, **altered motility** of the gastrointestinal tract, and **psychological factors**.

Based on the clinical manifestations, IBS can be classified as subtypes:

- **IBS-D** (Diarrhea predominant type)
- **IBS-C** (constipation-predominant type)
- **IBS-M**, that includes features of both IBS-D and C.

History findings typically include bowel habit changes for **at least 6 months**, which are usually **related to diarrhea or constipation**.

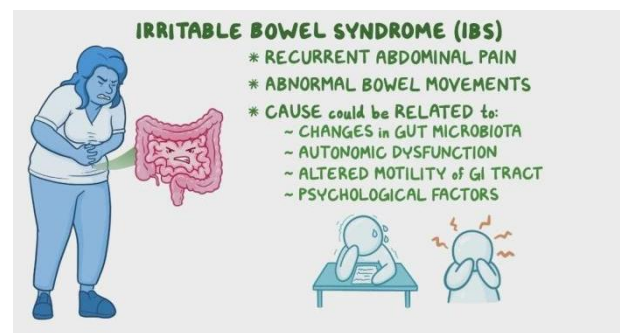
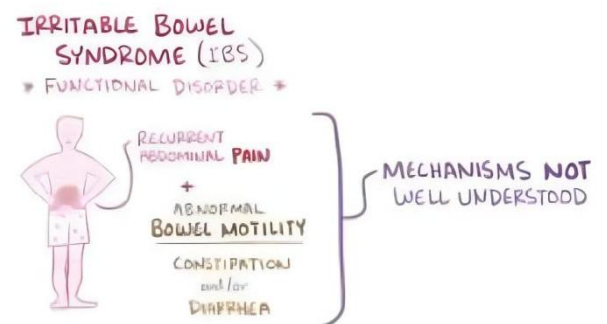
The patient will also report **abdominal pain** or discomfort that's typically **relieved with defecation**.

In some cases, **bloating**, or history of **depression**, **anxiety**, **fibromyalgia**, **trauma**, or recent infectious **gastroenteritis**.

On the physical exam, **abdominal tenderness** and **distention** could be elicited.

Before diagnosis of IBS , we should exclude the presence of **red flag features** , these includes:

- Hematochezia
- Unintentional weight loss
- A family history of colon cancer
- Acute onset of constipation in an older adult
- Change in stool caliber
- Anemia
- The presence of a rectal mass.



If the **red flag features** are present , we should consider **alternative diagnosis**.

The next step is to order labs to rule out other non-malignant gastrointestinal conditions with similar clinical manifestations , so the requested investigation will be:

- **CBC** to assess for anemia or infection
- Inflammatory markers like **ESR** and **CRP** to look for inflammation
- **TSH** to assess for thyroid dysregulation
- **Tissue transglutaminase IgA** for patient presenting with diarrhea to evaluate for possible Celiac disease.
- **Fecal calprotectin** to rule out colonic inflammation which may indicate uncontrolled inflammatory bowel disease.

Diagnosis of IBS:

we should assess for **Rome IV Criteria**.

Which define IBS as **recurrent abdominal pain** that occurs at least **1 day per week** in the **last 3 months**, with at least two of the following criteria: the pain is **related to defecation**, a change in stool **frequency**, or a change in **stool form**.

Management of IBS:

Diet modification, which includes consuming soluble fiber, and a trial of a diet low in a specific set of carbohydrates, which is also known as the low FODMAP diet.

Some examples of foods that the patient should limit include potatoes, brown rice, oats, and almonds.

The use of **peppermint oil**, has been shown to relax smooth muscle and target spasming of the GI tract.

As well as **tricyclic antidepressants**, like amitriptyline or nortriptyline, which are beneficial in relieving IBS associated abdominal pain.

Finally, **gut-directed psychotherapy** can be beneficial since IBS has a strong association with mental health.

Probiotics are not prescribed as there is a lack of evidence supporting their efficacy .

For patients with the **diarrhea-predominant type**, the goal is to increase stool firmness or decrease bowel movement frequency.

Treatment can be achieved by the **non-absorbed antibiotic rifaximin**, which works by addressing bacterial growth associated with diarrhea.

On the other hand, there's the **constipation-predominant type**.

Treatment for these individuals is geared towards stool softening and increasing bowel movement frequency.

There are two options available for treatment.

The first one includes **chloride channel activators, like lubiprostone**.

While the second one covers **guanylate cyclase activators, like linaclotide** which help soften the stool and stimulate bowel movement.

Inflammatory bowel disease (IBD)

Definition

Chronic inflammatory diseases of the gastrointestinal tract.

They are diagnosed by a set of **clinical, endoscopic, and histologic characteristics**, but no single finding is absolutely diagnostic for one disease or the other.

Moreover, some patients have a clinical picture that falls between the two diseases and are said to have **indeterminate colitis**.

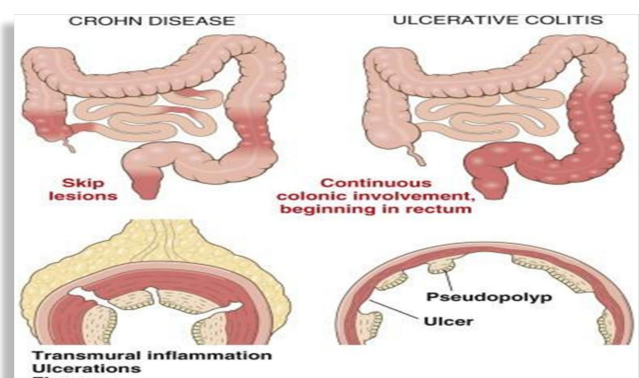
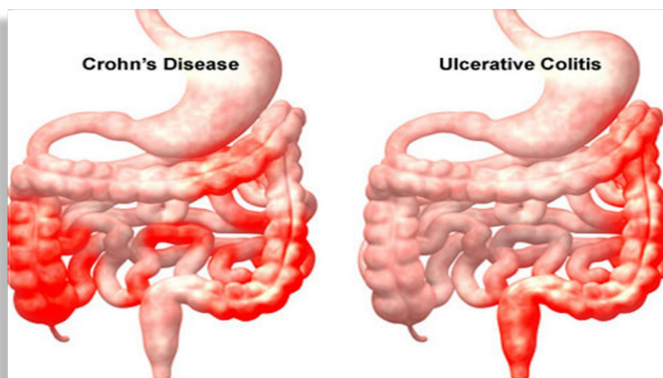
Epidemiology

The highest rates occur in white populations in northern Europe and North America, where the incidence for each disease is about 5 per 100,000 and the prevalence is approximately 50 per 100,000.

The peak age at onset is between 15 and 25 years of age, with a second, lesser peak between 55 and 65 years.

Pathology

Feature	Ulcerative Colitis	Crohn's Disease
<u>PATHOLOGY</u>		
Rectal involvement	Always	Common
Skip lesions	Never	Common
Transmural involvement	Rare	Common
Granulomas	Occasional	Common
Perianal disease	Never	Common
Cobblestone mucosa	Rare	Common
<u>RADIOLOGY</u>		
"Collar button" ulcers	Common	Occasional
Small intestinal involvement	Never	Common
Discontinuous involvement	Never	Common
Fistulas	Never	Common
Strictures	Occasional	Common
<u>ENDOSCOPY</u>		
Aphthous ulcers	Never	Common
Discontinuous involvement	Never	Common
Rectal sparing	Never	Common
Linear or serpiginous ulcers	Never	Common
Ulcers in the terminal ileum	Never	Common



Clinical features

Ulcerative Colitis

The dominant symptom in ulcerative colitis is **diarrhea**, which is usually associated with **blood in the stool**.

Other symptoms include **fever and pain**, which may be in either lower quadrant or in the rectum.

The **initial attack** of ulcerative colitis may be **Fulminant** with bloody diarrhea, but more commonly the disease begins indolently, with non-bloody diarrhea progressing to bloody diarrhea.

Crohn's Disease

Crohn's disease is marked by one of three major patterns:

- (1) disease in the **ileum and cecum (40% of patients)**.
- (2) disease confined to the **small intestine (30%)**.
- (3) disease confined to the **colon (25%)**.

The predominant symptoms are **diarrhea, abdominal pain, and weight loss**.

In patients with colonic disease, especially with rectal involvement, diarrhea is of small volume and associated with **urgency and tenesmus**.

Prolonged inflammation and scarring in the **rectum** can leave it so rigid and nondistensible that the patient is **incontinent**.

In disease confined to the **small intestine**, stools are of **larger volume** and not associated with urgency or tenesmus.

Patients with severe involvement of the **terminal ileum** and those who have undergone surgical resection of the terminal ileum may have bile salt diarrhea or **steatorrhea**.

Feature	Crohn's Colitis	Ulcerative Colitis
Mucosal lesions	<u>Aphthous ulcers</u> are common in early disease; late disease is notable for stellate, rake, bear-claw, linear, or serpiginous ulcers and <u>cobblestoning</u> .	Micro-ulcers are more common, but larger ulcers are seen <u>Pseudopolyps</u> are more common
Distribution	Often discontinuous and asymmetric, with <u>skipped segments</u> of normal intervening mucosa, especially in early disease	<u>Continuous, symmetric, and diffuse</u> , with granularity or ulceration found in the entirety of involved segments; however, peri-appendiceal inflammation (cecal patch) is common, even when the cecum is not involved
Rectum	<u>Complete, or more often relative, rectal sparing</u>	<u>Typically involved</u> with variable proximal distribution
Ileum	<u>Often involved</u> (=75% of cases)	<u>Not involved</u> , except as backwash ileitis in panulcerative colitis
Depth of inflammation	<u>Mucosal, submucosal, and transmural</u>	<u>Mucosal</u> ; transmural only in fulminant disease
Serosal findings	Marked erythema and creeping fat (the latter is virtually pathognomonic)	Absent except in severe colitis or toxic megacolon
Perianal complications	<u>Often prominent</u> , including large anal skin tags, deep fissures, <u>perianal fistulas</u> , that are often complex	<u>Not prominent</u> (fissure or fistula if present, should be uncomplicated)
Strictures	<u>Often present</u>	<u>Rarely present</u> ; when present, suggests adenocarcinoma
Fistulas	<u>Perianal, enterocutaneous, rectovaginal, enterovesicular, and other fistulas</u> may be present	<u>Absent</u> , except for the rare occurrence of rectovaginal or perianal fistula
Histopathology	<u>Granulomas are present in 15%-60% of patients</u> (higher frequency in surgical specimens than in mucosal pinch biopsies) Crypt abscesses may be present Focally enhanced inflammation, often on a normal background, is the hallmark	Granulomas should not be present (microgranulomas may be associated with ruptured crypt abscess) <u>Crypt abscesses and ulcers are the defining lesion</u> Ulceration on a background of inflamed mucosa
Serology	pANCA in 20%-25%, <u>ASCA in 41%-76%</u>	<u>pANCA in 60%-65%</u> , ASCA in 5%

Extra-intestinal Manifestations

COMPLICATIONS	CROHN DISEASE	ULCERATIVE COLITIS
Ocular disorders (uveitis, episcleritis)	+	+
Arthropathy	+	+
Oral ulcers	+	-
Skin disorders (pyoderma gangrenosum, erythema nodosum)	+	+
Nephrolithiasis	+	+
Primary sclerosing cholangitis	+	+
Bone disorders (osteoporosis, osteomalacia)	+	-
Thromboembolic disease	+	+
B ₁₂ deficiency	+	-

Diagnosis

Laboratory Studies

A **complete blood count (CBC)**, urinalysis, stool analysis and culture and serum chemistry tests are appropriate during the initial evaluation.

Serologic examination of blood for specific antibodies, perinuclear antineutrophil cytoplasmic antibodies (**pANCA**) [frequent in UC], antibodies to *Saccharomyces cerevisiae* (**ASCA**) [frequent in CD].

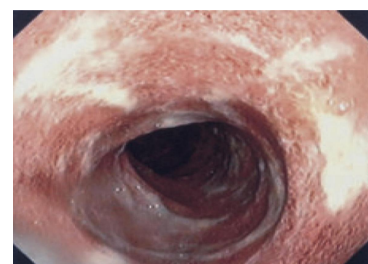
A newly recognized and clinically useful inflammatory marker for UC disease activity is **fecal calprotectin**, which is a protein secreted by neutrophils in the feces and is therefore a marker of intestinal inflammation.

Barium enema and barium follow through

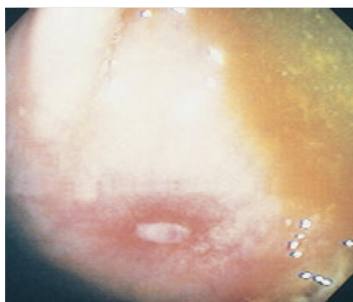
Endoscopy

Colonoscopy of ulcerative colitis

Friable mucosa, extensive ulceration, and exudates were detected.



Colonoscopy of Crohn's disease



A single aphthous ulcer, the earliest endoscopic finding in Crohn's disease



Multiple edematous inflammatory polyps give a "cobblestone" appearance to the mucosa.

Differential diagnosis

Differential Diagnosis of Inflammatory Bowel Disease

<u>Bacterial colitis</u>	<u>Behçet's colitis</u>
<i>Campylobacter</i>	<u>Sexually transmitted colitis</u>
<i>Shigella</i>	<i>Gonococcus</i>
<i>Salmonella</i>	<i>Chlamydia</i>
<i>Escherichia coli</i> (invasive)	Herpes
<i>Clostridium difficile</i> -associated colitis	Trauma
<u>Parasitic colitis</u>	<u>Crohn's disease look-alikes</u>
Amebiasis	Lymphoma
Schistosomiasis	<i>Yersinia</i>
<u>Ischemic colitis</u>	Tuberculosis
<u>Radiation colitis</u>	

Diagnosis	Clinical Features	Radiologic and Colonoscopic Features	Histologic Features
UC	<u>Bloody diarrhea</u>	Extends proximally from rectum; fine mucosal ulceration	Distortion of crypts; acute and chronic diffuse inflammatory cell infiltrate; goblet cell depletion; crypt abscesses; lymphoid aggregates
Crohn's colitis	<u>Perianal lesions are common; may be associated with ileitis; frank bleeding is less common than in UC</u>	Segmental disease; rectal sparing; strictures, fissures, ulcers, fistulas; small bowel involvement	Focal inflammation; submucosal involvement; granulomas; goblet cell preservation; transmural inflammation; fissuring
Ischemic colitis	<u>Occurs in older adults; sudden onset, often painful; usually resolves spontaneously in several days</u>	Segmental splenic flexure and sigmoid involvement are most common, with thumbprinting early and ulceration after 24-72 hr; rectal involvement is rare	Mucosal necrosis with ghost cells; congestion with red blood cells; hemosiderin-laden macrophages and fibrosis (when disease is chronic)
Microscopic colitis	<u>Watery diarrhea; normal-appearing mucosa at colonoscopy</u>	Usually normal	Chronic inflammatory infiltrate; increased intraepithelial lymphocytes (lymphocytic colitis) and/or subepithelial collagen band (collagenous colitis)
Infectious colitis	<u>Sudden onset; identifiable source in some cases (e.g., <i>Salmonella</i> spp.); pain may predominate (e.g., <i>Campylobacter</i> spp.); pathogens are present in stool</u>	Nonspecific findings	Crypt architecture is usually normal; edema, superficial neutrophilic infiltrate, crypt abscesses
Amebic colitis	History of travel to endemic area; amebae may be detected in a fresh stool specimen but ELISA for amebic lectin antigen is the preferable diagnostic test	Discrete ulcers; ameboma or strictures	Similar to UC; amebae present in lamina propria or in flask-shaped ulcers, identified by periodic acid-Schiff stain
Gonococcal proctitis	Rectal pain; pus	Granular changes in rectum	Intense polymorphonuclear neutrophil infiltration; purulent exudate; Gram-negative diplococci
Pseudomembranous colitis	<u>Often a history of antibiotic use; characteristic pseudomembranes may be seen on sigmoidoscopy; <i>Clostridium difficile</i> toxin is detectable in stools</u>	Edematous; shaggy outline of colon; pseudomembranes may be identified radiologically or seen at colonoscopy	May resemble acute ischemic colitis; summit lesions of fibrinopurulent exudate

Treatment

Diet and nutrition

Patients with mild symptomatic IBD usually are able to take food orally.

The diet should be **nutritious**.

Patients with Crohn's disease who have terminal ileal involvement and steatorrhea may require **supplemental fat-soluble vitamins, medium-chain triglycerides, and parenteral vitamin B12**.

Replacement **iron** may be indicated in patients who are iron-deficient.

Drugs

- Sulfasalazine and amino
- salicylates
- Corticosteroids.
- Antibiotics
- Immunomodulators
- Biologic therapy
- Probiotics and prebiotics

DRUG	DOSE	RELEASE SITE
5-AMINOSALICYLATES		
Sulfasalazine (Azulfidine)	2-6 g/day	Colon
Mesalamine (Asacol, Lialda, Apriso)	2.4-4.8 g/day	Distal ileum, colon
Olsalazine (Dipentum)	1-3 g/day	Colon
Balsalazide (Colazal)	6.25 g/day	Colon
Mesalamine (Pentasa)	2-4 g/day	Duodenum, jejunum, ileum, colon
Mesalamine (Rowasa), enema, suppository	4 g/day (enema) 1 g/day (suppository)	Rectum/sigmoid Rectum
Mesalamine (Canasa), suppository	1 g/day (suppository)	Rectum
CORTICOSTEROIDS		
Budesonide (Entocort EC)	Induction: 9 mg PO daily Maintenance: 6 mg PO daily	Small intestine
Budesonide (MMX, UCERIS)	Induction 9 mg PO daily	Colon
Prednisone	0.25-0.75 mg/kg PO daily	Systemic
Methylprednisolone	40-60 mg IV daily	Systemic
IMMUNOMODULATORS		
6-Mercaptopurine	1.5 mg/kg/day	Systemic
Azathioprine	2.5 mg/kg/day	Systemic
Methotrexate	Induction: 25 mg SC weekly × 4 mo. Maintenance: 15-25 mg SC weekly	Systemic
Cyclosporine	2-4 mg/kg/day IV	Systemic
BIOLOGICS		
Infliximab	Induction: 5 mg/kg IV weeks 0, 2, 6 Maintenance: 5-10 mg/kg IV every 8 weeks	Systemic
Adalimumab	Induction: 160 mg SC week 0, 80 mg week 2 Maintenance: 40 mg SC every other week	Systemic
Golimumab	Induction: 200 mg SC week 0 and 100 mg week 2 Maintenance: 100 mg SC every 4 weeks	Systemic
Certolizumab pegol	Induction: 400 mg SC weeks 0, 2, 4 Maintenance: 400 mg SC every 4 weeks	Systemic
Natalizumab	300 mg IV every 4 weeks	Systemic
Vedolizumab	Induction: 300 mg IV at 0, 2, and 6 weeks Maintenance: 300 mg IV every 8 weeks	Systemic

Corticosteroids

Historically, corticosteroids have been used in patients with severe UC or Crohn's disease to **induce a remission**.

Intravenous (IV) steroids (i.e., hydrocortisone 100 mg IV q6h or methyl prednisolone 10-30 mg IV q6-8h) are usually used in such patients.

When patients can take oral medications, **prednisone at doses of 40 to 60 mg per day** is usually given.

Antibiotics

Bacteria is known to play an **important role in the pathogenesis** of Crohn's disease and it may play a role in UC.

In Crohn's diseases, the indications for the use of antibiotics include **perianal disease**, **localized peritonitis** due to microperforation, **bacterial overgrowth** secondary to a chronic stricture, and as an **adjunct to drainage therapy** for abscesses and fistulas.

Metronidazole, Ciprofloxacin and Rifaximin can be used.

Immunomodulators and biologic therapy

Immunomodulator drugs act by **blocking lymphocyte proliferation, activation**, or effector mechanisms.

Azathioprine, Methotrexate, Cyclosporin can be used.

Anti-Tumor Necrosis Factor Antibody (biologic therapy)

TNF is a key inflammatory cytokine and mediator of intestinal inflammation. The expression of TNF is increased in IBD.

Infliximab is a chimeric mouse-human monoclonal antibody against TNF that is effective in CD.

Novel Therapies

Anti-Adhesion Molecules

Adhesion molecules are important in cellular trafficking in IBD and other diseases, in which immune and inflammatory cells from the periphery are recruited into sites of inflammation.

Gut-selective anti-adhesion molecule, **vedolizumab** shows promising results.

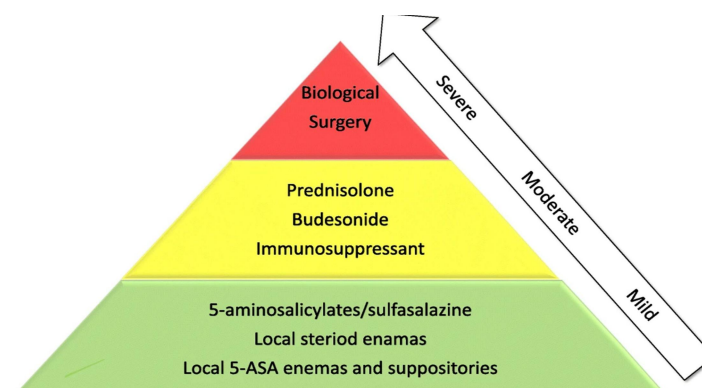
Kinase Inhibitors

The **Janus kinase (JAK) family of kinases**, which include JAK1 and JAK3 which are critical for **lymphocyte proliferation**, function, and activation.

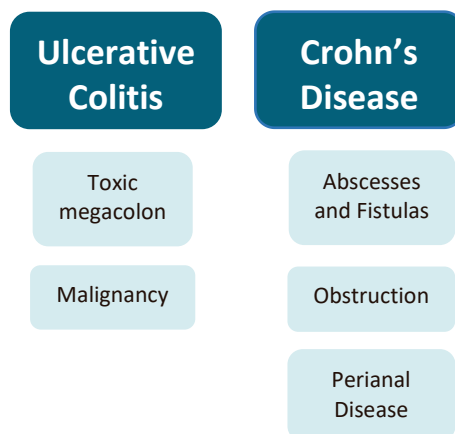
Tofacitinib, a novel oral inhibitor of JAK1 and JAK3, was recently shown to be safe and effective for the treatment of moderate to severe active UC.

Indications for surgical intervention

Ulcerative Colitis	Crohn's Disease
Intractable disease	Small Intestine
Fulminant disease	Stricture and obstruction unresponsive to medical therapy
Toxic megacolon	Massive hemorrhage
Colonic perforation	Refractory fistula
Massive colonic hemorrhage	Abscess
Extracolonic disease	Colon and Rectum
Colonic obstruction	Intractable disease
Colon cancer prophylaxis	Fulminant disease
Colon dysplasia or cancer	Perianal disease unresponsive to medical therapy
	Refractory fistula
	Colonic obstruction
	Cancer prophylaxis
	Colon dysplasia or cancer



Complications



Toxic megacolon

Toxic megacolon is a condition in which the colon **becomes atonic and dilated because of transmural inflammation**.

It is characteristically associated with severe ulcerative colitis, but it may complicate any severe inflammatory condition of the bowel, including **Crohn's disease, bacterial colitis**, amoebiasis, **pseudomembranous colitis**, and **ischemic colitis**.

Worsening of the patient's clinical condition and the development of **fever, tachycardia, and leukocytosis**.

Medical therapy is designed to reduce the likelihood of perforation.

If the patient does not begin to show signs of clinical improvement during the **first 24 to 48 hours** of medical therapy, the risk for perforation increases markedly, and **surgical intervention is indicated**.

Follow up for Malignancy

Patients with extensive ulcerative colitis have a markedly increased risk for **colon cancer** in comparison to the general population beginning **8 to 10 years after diagnosis** and increasing with time.

There should be **surveillance colonoscopy** with random biopsies in patients with long standing ulcerative colitis beginning 8 to 10 years after the onset of disease and **repeated every 1 to 2 years**.

If the specimens show **dysplasia the patient is sent for colectomy**.